
Prediction, Engineering, and Design of Protein Structures

17

Over a period of more than 3 billion years, a large variety of protein molecules has evolved to become the complex machinery of present-day cells and organisms. These molecules have evolved by random changes of genes by point mutations, exon shuffling, recombination and gene transfer between species, in combination with natural selection for those gene products that have conferred some functional advantage contributing to the survival of individual organisms.

Long before Darwin and Wallace proposed the theory of evolution and Mendel discovered the laws of genetics, plant and animal breeders had begun to interfere with the process of evolution in the species that gave rise to domesticated animals and cultivated plants. Considering their total lack of knowledge of both evolutionary theory and genetics, their achievements, brought about by forcing the pace of and subverting natural selection, were impressive albeit very gradual. With the advent of molecular genetics and in particular techniques for gene manipulation, we have now entered an era of genetic exploitation of organisms undreamed of only 50 years ago. We can now design genes to produce, in host organisms, novel gene products for the benefit of human beings; we are no longer restricted to selecting useful genes that arise by mutation. We are, however, only at the beginning of this new era, and so far we have only scratched the surface of the knowledge that is required for true engineering and design of protein molecules. We distinguish **protein engineering**, by which we mean mutating the gene of an existing protein in an attempt to alter its function in a predictable way, from **protein design**, which has the more ambitious goal of designing *de novo* a protein to fulfill a desired function.

Genome projects have now provided us with a description of the complete sequences of all the genes in more than a dozen organisms, and they will provide many more complete genome sequences within the next decade, including that of the human genome. These databases provide great opportunities for the analysis and exploitation of genes and their corresponding proteins. Central to reaping the intellectual and commercial benefits of this genetic information is the ability to find out the function of individual gene products. Almost all functional assignments to date have been based on sequence similarity to proteins of known function.

Knowledge of a protein's tertiary structure is a prerequisite for the proper understanding and engineering of its function. Unfortunately, in spite of recent significant technological advances, the experimental determination

of tertiary structure is still slow compared with the rate of accumulation of amino acid sequence data. This makes the **folding problem**, the successful prediction of a protein's tertiary structure from its amino acid sequence, central to rapid progress in post-genomic biology. We will, therefore, in this chapter first briefly describe implications of protein homology and methods for the prediction of secondary and tertiary structure before giving some examples of protein engineering and protein design.

Homologous proteins have similar structure and function

The term **homology** as used in a biological context is defined as similarity of structure, physiology, development and evolution of organisms based upon common genetic factors. The statement that two proteins are homologous therefore implies that their genes have evolved from a common ancestral gene.

Homologous proteins are mostly recognized by statistically significant similarities in their amino acid sequences. Usually, they also have similar functions although there are some known exceptions, where genes for ancient enzymes have been recruited at a later stage in evolution to produce proteins with quite different functions. An example is provided by one of the structural components in the eye lens that is homologous to the ancient glycolytic enzyme lactate dehydrogenase. Once a novel gene has been cloned and sequenced, a search for amino acid sequence similarity between the corresponding protein and other known protein sequences should be made. Usually, this is done by comparison with databases of known protein sequences using one of the standard sequence alignment computer programs.

Two proteins are considered to be homologous when they have identical amino acid residues in a significant number of sequential positions along the polypeptide chains. Using statistical methods based on comparisons of computer-generated random sequences, it is relatively straightforward to assess how many positions need to be identical for a statistically significant identity between two sequences. However, it is frequently found that two proteins with sequence identity below the level of statistical significance have similar functions and similar three-dimensional structures. In these cases, functionally important residues are identical and usually such residues form sequence patterns or motifs that can be used to identify other proteins that belong to the same functional family. Frequently, members of such families are also considered to be homologous, even though the identities are not statistically significant, only functionally significant. Databases for such families, based on identical or similar sequence motifs, are available on the World Wide Web (see pp. 393–394) and they are very useful for assigning function to a novel protein.

If significant amino acid sequence identity is found with a protein of known crystal structure, a three-dimensional model of the novel protein can be constructed, using computer modeling, on the basis of the sequence alignment and the known three-dimensional structure. This model can then serve as an excellent basis for identifying amino acid residues involved in the active site or in antigenic epitopes, and the model can be used for protein engineering, drug design, or immunological studies.

Since the sequence databases are large and growing exponentially, currently comprising more than 500,000 known protein sequences, the standard sequence alignment programs have been designed to provide a compromise between the speed and the accuracy of the search. As a result, they work well only when there is a reasonably high degree of sequence identity, usually of the order of 30% or more. Much more sensitive programs have been written that search for both identity and conserved structural properties and also for relatedness in different physical properties, but these inevitably require far more computing time. Carefully used, such programs can identify structural and functional similarity where the standard programs fail to do so.

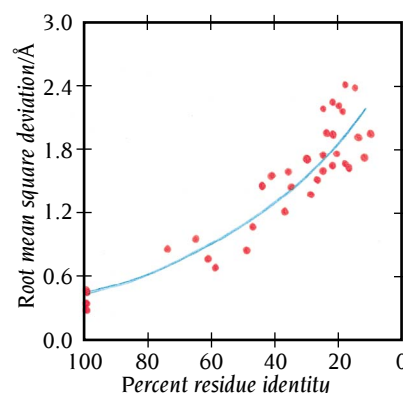
Homologous proteins have conserved structural cores and variable loop regions

Homologous proteins always contain a core region where the general folds of the polypeptide chains are very similar. This core region contains mainly the secondary structure elements that build up the interior of the protein: in other words, the scaffolds of homologous proteins have similar three-dimensional structures. Even distantly related proteins with low sequence identity have similar scaffold structures, although minor adjustments occur in the positions of the secondary structure elements to accommodate differences in the arrangements of the hydrophobic side chains in the interior of the protein. The greater the sequence identity, the more closely related are the scaffold structures (Figure 17.1). This has important implications for model building of homologous proteins; the more distantly related two proteins are, the more the scaffold must be adjusted to model the new structure.

Loop regions that connect the building blocks of scaffolds can vary considerably both in length and in structure. The problem of predicting the three-dimensional structure of a protein that is homologous to a protein of known three-dimensional structure is therefore mainly a question of predicting the structure of loop regions and side-chain conformations, after the scaffold has been adjusted. As mentioned in Chapter 2, loop regions do not have random structures, and their main-chain conformations cluster in sets of similar structures. The conformation of each set depends more on the number of amino acids in the loop and the type of secondary structure elements that it connects, whether they are α - α , β - β , α - β , or β - α connections, than on the actual amino acid sequences. Therefore it is possible to use a database of loop regions from proteins of known structure to obtain a preliminary model of the loops of an unknown structure. To model a protein structure, suitable main-chain loop conformations from this database are attached to the scaffold modeled to have a structure similar to that of the known homologous protein. Finally, the conformations of the side chains are predicted by energy refinement of the model, which minimizes the free energy of the protein by maximizing the interaction energies of the amino acids. Analysis of structures determined to high resolution has shown that only a few side-chain conformations frequently occur. These are called rotamers and model building of side chains employs databases of such rotamers.

An instructive example of the use of such procedures has been in modeling antigen-binding sites in immunoglobulins. These binding sites are built up from three hypervariable loop regions, CDR1–CDR3, from the variable domains of both the light and the heavy chains of immunoglobulins as described in Chapter 15. There is usually high sequence identity within the scaffolds of the variable domains in different immunoglobulin molecules. Consequently, the scaffold of variable domains of known three-dimensional structures can be used in modeling a new monoclonal antibody with a known amino acid sequence. However, the CDR regions of a new antibody are usually very different in sequence from those of any other known antibody, and their three-dimensional structures must be predicted. By comparing

Figure 17.1 The relation between the divergence of amino acid sequence and three-dimensional structure of the core region of homologous proteins. Known structures of 32 pairs of homologous proteins such as globins, serine proteinases, and immunoglobulin domains have been compared. The root mean square deviation of the main-chain atoms of the core regions is plotted as a function of amino acid homology (red dots). The curve represents the best fit of the dots to an exponential function. Pairs with high sequence homology are almost identical in three-dimensional structure, whereas deviations in atomic positions for pairs of low homology are of the order of 2 Å. (From C. Chothia and A. Lesk, *EMBO J.* 5: 823–826, 1986.)



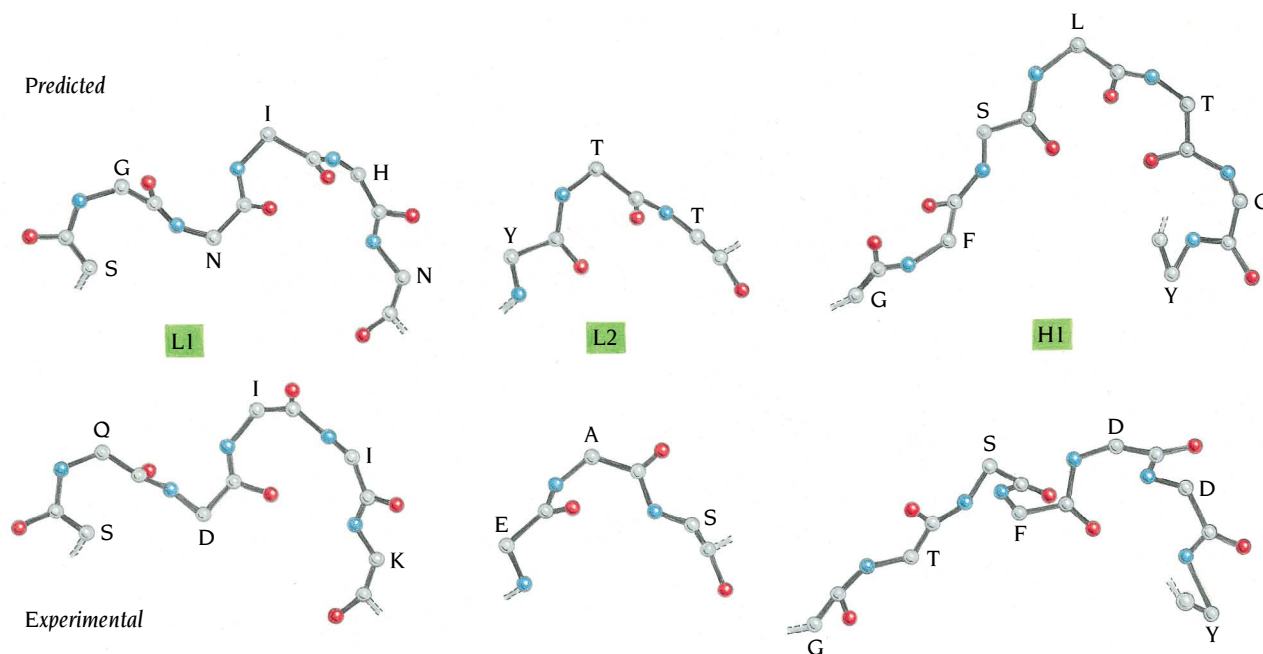


Figure 17.2 An example of prediction of the conformations of three CDR regions of a monoclonal antibody (*top row*) compared with the unrefined x-ray structure (*bottom row*). L1 and L2 are CDR regions of the light chain, and H1 is from the heavy chain. The amino acid sequences of the loop regions were modeled by comparison with the sequences of loop regions selected from a database of known antibody structures. The three-dimensional structure of two of the loop regions, L1 and L2, were in good agreement with the preliminary x-ray structure, whereas H1 was not. However, during later refinement of the x-ray structure errors were found in the conformations of H1, and in the refined x-ray structure this loop was found to agree with the predicted conformations. In fact, all six loop conformations were correctly predicted in this case. (From C. Chothia et al., *Science* 233: 755–758, 1986.)

known antibody structures and sequences, it has been shown that there is only a small repertoire of main-chain conformations for at least five of the six CDR regions and that the particular conformation adopted is determined by a few key conserved residues for each loop conformation. For example, three different conformations were found for the CDR3 regions of the light chains in nine known x-ray structures. More than 90% of the known sequences of light-chain CDR3 regions obey the sequence constraints of one or other of these three conformations. By using this repertoire of loop conformations, considerable success has been achieved in correctly predicting the structure of antigen-binding surfaces. An example of such a prediction compared with the actual structure, subsequently determined, is given in Figure 17.2.

Knowledge of secondary structure is necessary for prediction of tertiary structure

What can be done by predictive methods if the sequence search fails to reveal any homology with a protein of known tertiary structure? Is it possible to model a tertiary structure from the amino acid sequence alone? There are no methods available today to do this and obtain a model detailed enough to be of any use, for example, in drug design and protein engineering. This is, however, a very active area of research and quite promising results are being obtained; in some cases it is possible to predict correctly the type of protein, α , β , or α/β , and even to derive approximations to the correct fold.

Today's predictive methods rely on prediction of secondary structure: in other words, which amino acid residues are α -helical and which are in β strands. We have emphasized in Chapter 12 that secondary structure cannot in general be predicted with a high degree of confidence with the possible exceptions of transmembrane helices and α -helical coiled coils. This imposes a basic limitation on the prediction of tertiary structure. Once the correct secondary structure is known, we know enough about the rules for packing elements of secondary structure against each other (see Chapter 2 for helix packing) to derive a very limited number of possible stable globular folds. Consequently, secondary structure prediction lies at the heart of the prediction of tertiary structure from the amino acid sequence.

Unfortunately for predictive methods, secondary and tertiary structures are closely linked in the sense that global tertiary structure imposes local secondary structure at least in some regions of the polypeptide chain. The ability of a specific short sequence of amino acids to form an α helix, a β strand, or a loop region is dependent not only on the sequence of that region but also on its environment in the three-dimensional structure. For example, by analyzing all the known tertiary structures, it has been shown that peptide regions of up to five residues long with identical amino acid sequences are α -helical in one structure and a β strand or a loop in other structures. While this interdependence of secondary and tertiary structure complicates secondary structure predictions, it can, sometimes, be used to improve such predictions, by an iterative scheme in which a preliminary assignment of secondary structure is used to predict the type of domain structure, for example, a four-helix bundle or an α/β barrel. The structure type of the domain imposes additional constraints on possible secondary structure, which can be used to refine the secondary structure prediction.

Prediction methods for secondary structure benefit from multiple alignment of homologous proteins

Over 20 different methods have been proposed for predictions of secondary structure; they can be categorized in two broad classes. The empirical statistical methods use parameters obtained from analyses of known sequences and tertiary structures. All such methods are based on the assumption that the local sequence in a short region of the polypeptide chain determines local structure; as we have seen, this is not a universally valid assumption. The second group of methods is based on stereochemical criteria, such as compactness of form with a tightly packed hydrophobic core and a polar surface. Three frequently used methods are the empirical approaches of P.Y. Chou and G.D. Fasman and of J. Garnier, D.J. Osguthorpe and B. Robson (the GOR method), and third, the stereochemical method of V.I. Lim.

Although these three methods use quite different approaches to the problem, the accuracy of their secondary structure prediction is about the same. All three methods can be used to assign one of three states to each residue: α helix, β strand, or loop. Random assignment of these three states to residues in a polypeptide chain will give an average score of 33% correctly predicted states. The methods have been assessed in an analysis of single sequences of a large number of known x-ray structures comprising more than 10,000 residues. For the three-state definition of secondary structure, the overall accuracy of prediction was about 55%. Other objective assessments have given similar results.

However, when these predictive methods are used on a set of homologous proteins the predictive power is considerably higher. The underlying assumption is that secondary and tertiary structure has been more conserved during evolution than amino acid sequence; in other words only such changes have been retained during evolution that conserve the structure. Consequently, the pattern of residue changes within homologous proteins contains specific information about the structure. Conserved hydrophobic residues are usually in the interior of the protein with a high probability of belonging to helices or sheet strands. Insertions and deletions almost always occur in loop regions and not in the scaffold built up from helices and strands.

Several programs are now available that use multiple alignment of homologous proteins for prediction of secondary structure. One such program, called PHD, which was developed by Chris Sander and coworkers, EMBL, Heidelberg, has reached a mean accuracy of prediction of 72% for new structures.

A large fraction of the remaining errors occur at the ends of α helices and β strands and, in addition, some errors occur because of occasional difficulties

in distinguishing between α helices and β strands. These latter errors can be corrected if the structural class, α , β , or α/β , can be deduced from a combination of physical studies, for example, circular dichroism spectra, and the general features of the secondary structure prediction. For example, if the prediction scheme assigns one or two short α helices among many β strands in a protein of the β class, there is a high probability that the regions of secondary structures are essentially correctly predicted but that they should all be β strands.

These predictive methods are very useful in many contexts; for example, in the design of novel polypeptides for the identification of possible antigenic epitopes, in the analysis of common motifs in sequences that direct proteins into specific organelles (for instance, mitochondria), and to provide starting models for tertiary structure predictions.

Many different amino acid sequences give similar three-dimensional structures

How many completely different amino acid sequences might give a similar three-dimensional structure for an average-sized domain of 150 amino acid residues? Simple combinatorial calculations show that there are a total of 20^{150} or roughly 10^{200} possible amino acid sequences for such a domain, given the 20 different amino acids in natural proteins. This number is much larger than the number of atoms in the known universe. A more laborious calculation shows that out of these 10^{200} possible combinations we can extract about 10^{38} members that have less than 20% amino acid sequence identity with each other and that therefore can be considered to have different sequences. In other words, there are 10^{38} different ways of constructing a domain of 150 amino acids using the 20 standard amino acids as building blocks. We do not know how many of these can form a stable three-dimensional structure but, assuming say that one out of a billion (10^9) can, we are left with 10^{29} folded possible proteins. In the previous chapters we have seen that simple structural motifs arrange themselves into a limited number of topologically different domain structures. It has been estimated on reasonable grounds that there are about 1000 topologically different domain structures. Since there are 10^{29} possible different sequences that might fold into 10^3 different structures, it follows that there are of the order of 10^{26} different side chain arrangements with less than 20% amino acid sequence identity that can give similar polypeptide folds. Only a small fraction of these possible proteins will be found in nature.

For each of the 500 or so different domain structures that have so far been observed, we might at best know about a dozen of these different possible sequences. It is not trivial to recognize the general sequence patterns that are common to specific domain structures from such a limited knowledge base.

Prediction of protein structure from sequence is an unsolved problem

How to predict the three-dimensional structure of a protein from its amino acid sequence is the major unsolved problem in structural molecular biology. We would like to have a computer program that could simulate the action of the processes that operate in a test tube or a living cell when a polypeptide chain with a specific amino acid sequence folds into a precise three-dimensional structure. Why is this prediction of protein folding so difficult? The answer is usually formulated in terms of the complexity of the task of searching through all the possible conformations of a polypeptide chain to find those with low energy. It requires enormous amounts of computing time, in addition to the complication discussed in Chapter 6 that the energy difference between a stable folded molecule and its unfolded state is a small number containing large errors.

With the realization that there are only a limited number of stable folds and many unrelated sequences that have the same fold, biologically oriented computer scientists started to address what is called the **inverse folding problem**; namely, which sequence patterns are compatible with a specific fold? If this question can be answered, such patterns could be used to search through the genome sequence databases and extract those sequences that have a specific fold, such as the α/β barrel or the immunoglobulin fold.

However, given the large number of possible unrelated sequences for each fold and the limited number of known sequences, a variation of this problem has recently been addressed by a large number of groups; namely, which of the known folds, if any, is most compatible with a specific sequence? The methodology used is called **threading** because it involves threading a specific sequence through all known folds and, for each fold, estimating the probability that the sequence can have that fold. Considerable progress has recently been made in threading, and in blind tests several structures have been correctly predicted by different groups.

Threading methods can assign amino acid sequences to known three-dimensional folds

Threading methods, which are also called protein fold assignments or fold recognition, are a promising and rapidly evolving field of computational structural biology. The goal is to assign to each genome-derived protein sequence the protein fold to which it most closely corresponds, or to determine whether there is no known fold to which the sequence belongs. A further goal is to align the new sequence properly to the three-dimensional structure of the fold to which it belongs to provide a low-resolution model. In order to test different methods of threading, blind tests are arranged, called Critical Assessment of Structure Prediction (CASP), in which the participants are given sequences and invited to predict the fold and make an alignment before the structure is determined experimentally. We will briefly describe here the methods used by one of the more successful participants in these tests, the group of David Eisenberg at University of California, Los Angeles.

The first requirement for threading is to have a database of all the known different protein folds. Eisenberg has used his own library of about 800 folds, which represents a minimally redundant set of the more than 6000 structures deposited at the Protein Data Bank. Other groups use databases available on the World Wide Web, where the folds are hierarchically ordered according to structural and functional similarities, such as SCOP, designed by Alexey Murzin and Cyrus Chothia in Cambridge, UK.

For each fold one searches for the best alignment of the target sequence that would be compatible with the fold; the core should comprise hydrophobic residues and polar residues should be on the outside, predicted helical and strand regions should be aligned to corresponding secondary structure elements in the fold, and so on. In order to match a sequence alignment to a fold, Eisenberg developed a rapid method called the 3D profile method. The environment of each residue position in the known 3D structure is characterized on the basis of three properties: (1) the area of the side chain that is buried by other protein atoms, (2) the fraction of side chain area that is covered by polar atoms, and (3) the secondary structure, which is classified in three states: helix, sheet, and coil. The residue positions are rather arbitrarily divided into six classes by properties 1 and 2, which in combination with property 3 yields 18 environmental classes. This classification of environments enables a protein structure to be coded by a sequence in an 18-letter alphabet, in which each letter represents the environmental class of a residue position.

Each of the 20 different amino acids has different preferences for each of the 18 environmental classes; for instance a Leu has a high preference for being in a helical class with a high fraction of buried side chain area, whereas

an Asp has a very low preference for that position. Numerical values for these preferences, called 3D-1D scores, were derived from a set of well-refined high-resolution protein structures, together with sets of sequences similar to the sequences of the 3D structures. This produced a scoring table in which for each environmental class a numerical value of preference is associated with each of the 20 amino acids. This table is used to set up a 3D profile table of a protein structure, in which each residue position is assigned an environmental class with corresponding numerical values for preference for each type of amino acid. The essence of this method is that the three-dimensional structure is reduced to a one-dimensional array, which facilitates matching to a one-dimensional sequence.

A target amino acid sequence is aligned against this structure profile in such a way that the best possible match—the highest total score—is obtained, allowing gaps and insertions. Such an alignment is conceptually similar to alignment of two sequences and similar methods have been used. The match of a sequence to a 3D structure profile for a specific fold is expressed quantitatively by a value called the Z-score, which is the number of standard deviations above the mean alignment score for other sequences of similar length. A high Z-score means there is a high probability that the sequence has the corresponding fold.

The methods described here have subsequently been improved and extended by Eisenberg, but the principle remains essentially the same. Other groups use different methods to screen the sequence–structure alignments and different criteria to assess the matches. Manfred Sippl at the University of Salzburg, Austria, has developed a set of potentials to screen and assess the alignments, the essence of which is to maximize the number of hydrophobic interactions and to minimize the number of buried polar atoms that do not participate in hydrogen bonds. These and similar potentials are now used by many groups in their threading programs. Correct folds can be predicted with a reasonably high probability for small and medium-sized proteins. Correct alignment of the sequence to the selected fold is, however, less accurate.

Proteins can be made more stable by engineering

Protein engineering, via site-directed mutagenesis of DNA, can be used to answer very specific questions about protein stability, and the results of these studies are now being used to increase the stability of industrially important enzymes. To illustrate some of the factors of importance for protein stability that have been revealed by protein engineering studies, we have chosen the extensive work on the enzyme lysozyme from bacteriophage T4 that has been done by the group of Brian Mathews, University of Oregon, Eugene.

Lysozyme from bacteriophage T4 is a 164 amino acid polypeptide chain that folds into two domains (Figure 17.3) There are no disulfide bridges; the two cysteine residues in the amino acid sequence, Cys 54 and Cys 97, are far apart in the folded structure. The stability of both the wild-type and mutant proteins is expressed as the melting temperature, T_m , which is the temperature at which 50% of the enzyme is inactivated during reversible heat denaturation. For the wild-type T4 lysozyme the T_m is 41.9 °C.

We will discuss three different approaches to engineer a more thermostable protein than wild-type T4 lysozyme, namely (1) reducing the difference in entropy between folded and unfolded protein, which in practice means reducing the number of conformations in the unfolded state, (2) stabilizing the α helices, and (3) increasing the number of hydrophobic interactions in the interior core.

Disulfide bridges increase protein stability

The greater the number of unfolded conformations of a protein, the higher the entropic cost of folding that protein into its single native state (see Chapter 6). Reducing the number of unfolded conformations therefore increases

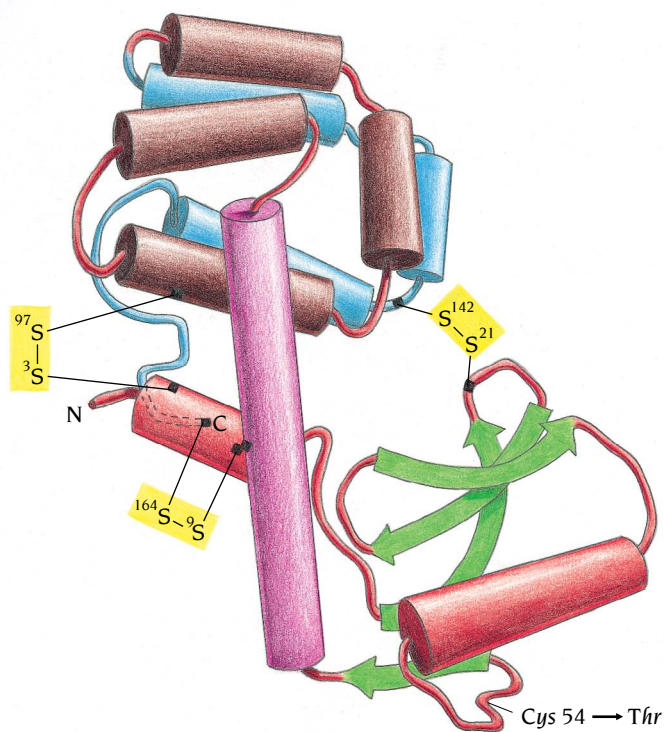


Figure 17.3 The polypeptide chain of lysozyme from bacteriophage T4 folds into two domains. The N-terminal domain is of the $\alpha + \beta$ type, built up from two α helices (red) and a four-stranded antiparallel β sheet (green). The C-terminal domain comprises seven short α helices (brown and blue) in a rather irregular arrangement. (The last half of this domain is colored blue for clarity.) One long α helix connects the two domains (purple). Thermostable mutants of this protein were constructed by introducing disulfide bridges at three different places (yellow). The position of Cys 54, which was mutated to Thr, is also shown. (Adapted from M. Matsumura et al., *Nature* 342: 291–293, 1989.)

the stability of the native state. The most obvious way to decrease the number of unfolded conformations is to introduce a novel disulfide bond based on knowledge of the tertiary structure of the folded protein. The longer the loop between the cysteine residues, the more restricted is the unfolded polypeptide chain, giving more stabilization of the folded structure. To design such bridges is, however, not a simple task, since the geometry of an unstrained $-\text{CH}_2-\text{S}-\text{S}-\text{CH}_2-$ bridge in proteins is confined to rather narrow conformational limits, and deviations from this geometry will introduce strains into the folded structure and hence reduce rather than increase its stability. It is, therefore, not sufficient to choose at random two residues close together in space to make such a bridge, rather the protein engineer must carefully select pairs of residues with main-chain conformations that fulfill the conditions needed for an unstrained disulfide bridge.

Mathews made a very careful comparison between the geometry of the 295 disulfide bridges in known x-ray structures and all possible pairs of amino acid residues close enough to each other in the refined T4 lysozyme structure to accommodate a disulfide bridge. This was followed by energy minimization of the most likely candidate disulfide bridges and an analysis of stabilizing interactions present in the wild-type structure that would be lost by mutation to a Cys residue. Such losses should be minimized. Three candidate disulfide bridges remained after this filtering, one of which, Cys 3–Cys 97, contained one of the cysteine residues (Cys 97) that is present in the wild type. The five amino acid residues—Ile 3, Ile 9, Thr 21, Thr 142, and Leu 164 (see Figure 17.3)—were mutated to Cys residues in separate experiments so that all single (3–97, 9–164, and 21–142) as well as combinations of double and triple disulfide bonds could be formed. In addition, the second Cys residue of the wild-type enzyme, Cys 54, was mutated to Thr to avoid the formation of incorrect disulfide bonds during folding.

The results of this careful design of novel disulfide bridges were very encouraging (Figure 17.4). All the mutants were more stable in their oxidized forms than wild-type protein. The longer the loop between the cysteine

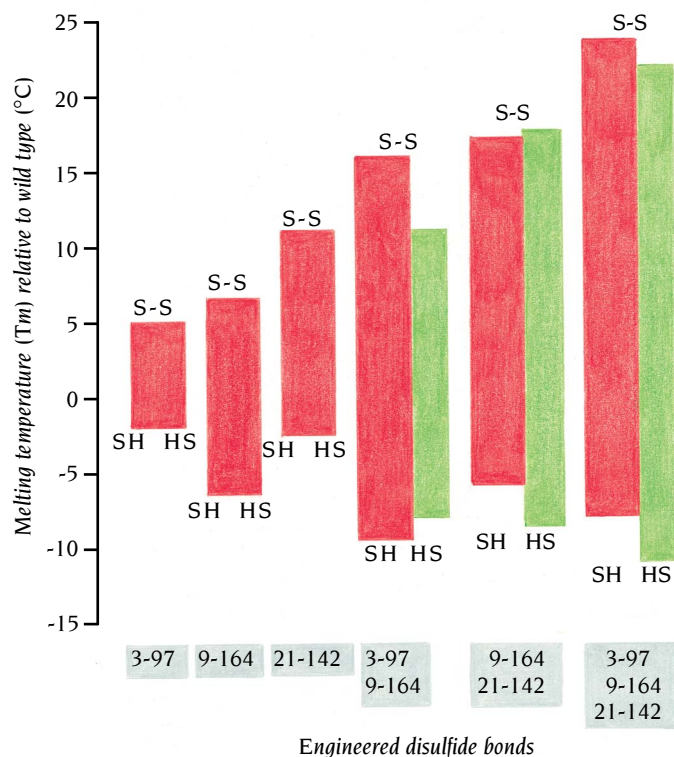


Figure 17.4 Melting temperatures, T_m , of engineered single-, double-, and triple-disulfide-containing mutants of T4 lysozyme relative to wild-type lysozyme. The red bars show the differences in T_m values of the oxidized and reduced forms of the mutant lysozymes. The green bars for the multiple-bridged proteins correspond to the sum of the differences in T_m values for the constituent single-bridged lysozymes. (Adapted from M. Matsumura et al., *Nature* 342: 291–293, 1989.)

residues of the mutants with single disulfide bonds, the larger was the effect on stability. Furthermore, the effects were additive so that the increase in T_m of 23 °C for the mutant with three disulfide bonds was approximately equal to that of the sum of the increases in T_m values for the three mutants with single disulfide bonds (4.8 °C + 6.4 °C + 11.0 °C = 22.2 °C). The effect on the stability of the protein from reducing the number of possible unfolded structures through introduction of disulfide bridges, the entropic effect, is even larger than these values show because the reduced forms of the mutants had a lower T_m than wild type, which indicates that favorable contacts in the folded structure had been lost by the mutations. These experiments show that engineered disulfide bridges can be combined together to enhance stability dramatically. Needless to say, knowledge of the three-dimensional structure of the protein is a prerequisite to engineer increased stability in this way.

Glycine and proline have opposite effects on stability

Glycine residues have more conformational freedom than any other amino acid, as discussed in Chapter 1. A glycine residue at a specific position in a protein has usually only one conformation in a folded structure but can have many different conformations in different unfolded structures of the same protein and thereby contribute to the diversity of unfolded conformations. Proline residues, on the other hand, have less conformational freedom in unfolded structures than any other residue since the proline side chain is fixed by an extra covalent bond to the main chain. Another way to decrease the number of possible unfolded structures of a protein, and hence stabilize the native structure, is, therefore, to mutate glycine residues to any other residue and to increase the number of proline residues. Such mutations can only be made at positions that neither change the conformation of the main chain in the folded structure nor introduce unfavorable, or cause the loss of favorable, contacts with neighboring side chains.

Both types of mutations have been made in T4 lysozyme. The chosen mutations were Gly 77→Ala, which caused an increase in T_m of 1 °C, and Ala 82→Pro, which increased T_m by 2 °C. The three-dimensional structures of these mutant enzymes were also determined: the Ala 82→Pro mutant had a structure essentially identical to the wild type except for the side chain of residue 82; this strongly indicates that the effect on T_m of Ala 82→Pro is indeed due to entropy changes. Such effects are expected to be additive, so even though each mutation makes only a small contribution to increased stability, the combined effect of a number of such mutations should significantly increase a protein's stability.

Stabilizing the dipoles of α helices increases stability

In Chapter 2 we described the α helix as a dipole with a positive charge at its N-terminus and a negative charge at the C-terminus. Negative ions, such as phosphate groups in coenzymes or substrates, are usually bound to the positive ends of such helical dipoles. The α helices that are not part of a binding site frequently have a negatively charged side chain at the N-terminus or a positively charged residue at the C-terminus that interacts with the dipole of the helix. Such dipole-compensating residues stabilize the helical forms of small synthetic peptides in solution. Do these helix-stabilizing residues also contribute to the overall stability of globular proteins? Of the 11 α helices of T4 lysozyme, 7 helices have negatively charged residues close to their N-termini; two of the remaining four α helices were therefore chosen for engineering studies to answer this question (Figure 17.5).

Two different mutant proteins with single substitutions at the N-terminus of each of these helices, Ser 38→Asp and Asn 144→Asp, were made as well as the corresponding double mutant. The single mutants both showed an increase in T_m of about 2 °C; the effects are additive since the double mutant had a T_m about 4 °C higher than wild type. This corresponds to 1.6 kcal/mol of stabilization energy. From the x-ray structures of these mutants it is apparent that the stabilization is due to electrostatic interactions and not to specific hydrogen bonding between the substituted amino acid and the end of the helix. Alan Fersht in Cambridge, UK has shown, using a different system, the small bacterial ribonuclease, barnase, that a histidine residue at

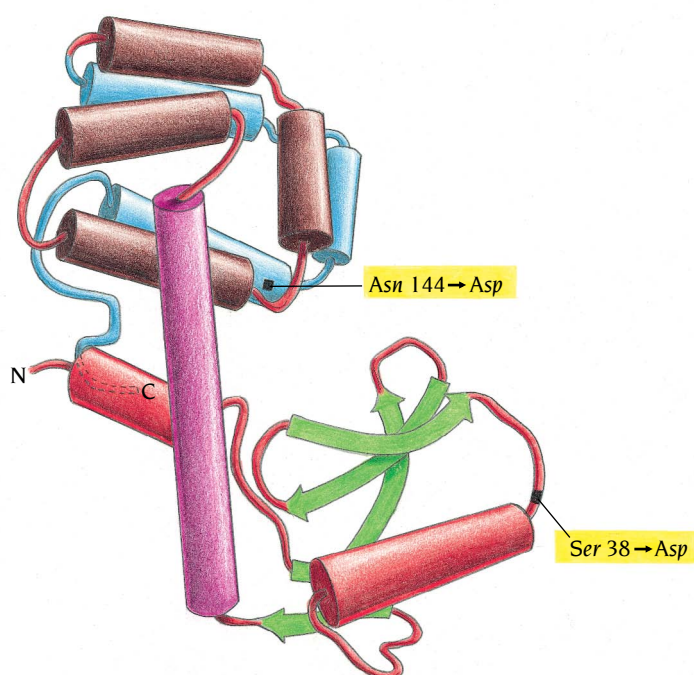


Figure 17.5 Diagram of the T4 lysozyme structure showing the locations of two mutations that stabilize the protein structure by providing electrostatic interactions with the dipoles of α helices. (Adapted from H. Nicholson et al., *Nature* 336: 651–656, 1988.)

the C-terminus of a helix stabilizes the barnase structure by about 2.1 kcal/mol. Significant stabilization of α -helical structures might, therefore, be obtained by combining several such helix-stabilizing mutations.

Mutants that fill cavities in hydrophobic cores do not stabilize T4 lysozyme

We emphasized in Chapter 2 that burying the hydrophobic side chains in the interior of the molecule, thereby shielding them from contact with solvent, is a major determinant in the folding of proteins. The surface that is buried inside a folded protein contributes directly to the stabilization energy of the molecule. Studies of destabilizing mutants in barnase, where **cavities** have been engineered into the hydrophobic core of the wild-type enzyme by mutations such as Ile to Val or Phe to Leu show that the introduction of a cavity the size of one $-\text{CH}_2-$ group destabilizes the enzyme by about 1 kcal/mol. By analogy it should be possible to stabilize a wild-type protein by making mutations that fill existing cavities in its hydrophobic core. Even though proteins have the atoms of their hydrophobic cores packed approximately as tight as atoms are packed in crystals of simple organic molecules, there are cavities in the cores of almost every protein.

T4 lysozyme has two such cavities in the hydrophobic core of its α helical domain. From a careful analysis of the side chains that form the walls of the cavities and from building models of different possible mutations, it was found that the best mutations to make would be Leu 133–Phe for one cavity and Ala 129–Val for the other. These specific mutants were chosen because the new side chains were hydrophobic and large enough to fill the cavities without making too close contacts with surrounding atoms.

The two single mutants were constructed, purified, analyzed for stability, and crystallized. They were both less stable than wild type by 0.5 to 1.0 kcal/mol. The x-ray structures of the mutants provide a rational explanation for this disappointing result. It turns out that in order to fill the cavities, the new side chains in the mutants adopt energetically unfavorable conformations. This introduces strain in the structure, which obviously costs more energy than is gained by the new hydrophobic interactions. Even careful model building is obviously not sufficient to predict detailed structural and energetic effects of mutations in the hydrophobic core of proteins. Apparently, the observed core structure in T4 lysozyme, and probably in most proteins, reflects a compromise between the hydrophobic effect, which will tend to maximize the core-packing density, and the strain energy that would be incurred in eliminating all packing defects. Therefore, mutations designed to fill existing cavities may be effective in some cases, but they are not likely to provide a general route to substantial improvement in protein stability.

Proteins can be engineered by combinatorial methods

The ultimate goal of protein engineering is to design proteins to carry out predicted functions. However, we do not yet completely understand the rules governing protein folding and molecular recognition, making design of proteins difficult. Protein engineers have therefore invented **combinatorial methods**, in which **libraries** of related proteins are analyzed simultaneously. By sorting these libraries to select for a particular function, the small number of active proteins can be separated from millions of inactive variants. Combinatorial libraries have been used to increase the activity of enzymes, to improve the binding affinity and specificity of proteins, and even to identify novel peptide ligands. Additionally, researchers hope to use the structural and functional data obtained through library selection to improve their ability precisely to engineer molecular interactions.

Combinatorial methods are often referred to as *in vitro* or directed evolution techniques. In nature, the random DNA mutations that lead to changes in protein sequences occur rarely and so evolution is usually a slow

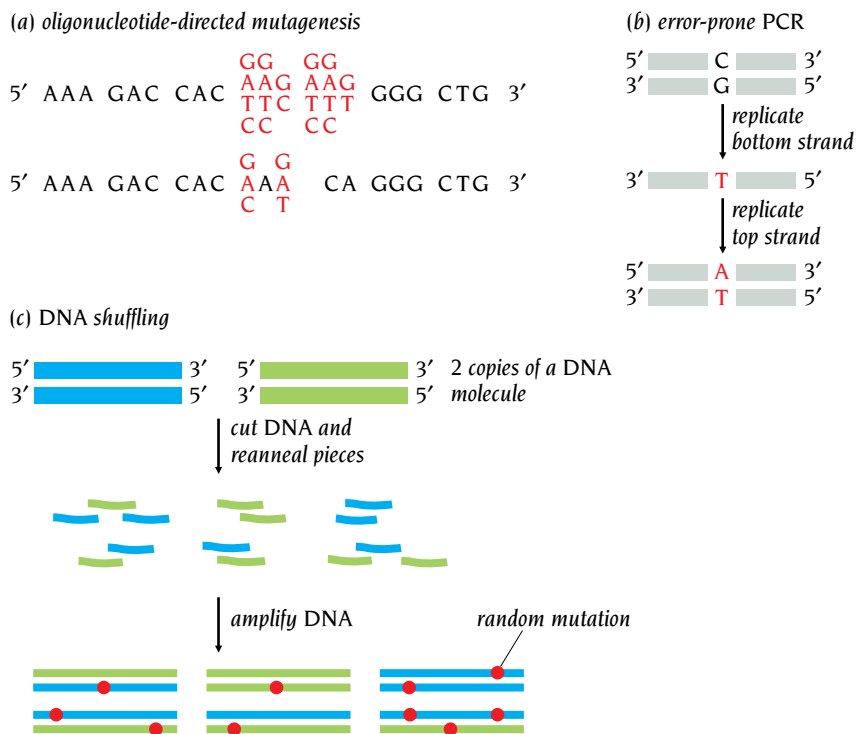


Figure 17.6 Methods of random mutagenesis. Several techniques are available for generating DNA libraries. (a) Oligonucleotides (short molecules of DNA) can be synthesized to contain mixtures of nucleotides at specific codons. An NNS or NNK codon, containing all bases at the first two positions and only two bases at the third position, each allows 20 possible amino acids. Alternatively, a restricted set of bases gives a more focused library; the lower example shown would permit only hydrophilic amino acids (E, K, Q, D, N, H). Oligonucleotide libraries are then incorporated into the gene of interest. (b) Error-prone polymerase chain reaction (PCR) uses a DNA polymerase to replicate the target gene. Conditions are chosen to decrease the fidelity of replication, leading to single base pair errors throughout the gene. (c) In DNA shuffling, a gene is first cut into pieces and then regenerated using a DNA polymerase. The polymerase introduces mutations similar to error-prone PCR; additionally, the pieces of DNA get mixed, so that mutations from separate copies of the gene can be combined.

process. Combinatorial methods accelerate evolution by controlling both the level and location of genetic mutation. Usually, a large number of mutations are concentrated in a single gene through **random mutagenesis**. Because mutagenesis techniques differ in the number and dispersion of mutations introduced to a gene, the appropriate method of mutagenesis depends on what questions the protein engineer seeks to address; the most widely used mutagenesis strategies are outlined in Figure 17.6. The mutated genes are then selected *in vivo*, by conferring a function to cells, or *in vitro*, by binding to an immobilized target. The most common method for *in vitro* selection, **phage display**, is discussed in the following section. The optimal strategies for generating and sorting a library depend on the affinity and specificity of the library for the target. The following examples will therefore illustrate some important combinatorial methods as well as the information that has been gained by using these techniques.

Phage display links the protein library to DNA

In designing a selection strategy, we must consider how to isolate and characterize the functional proteins in a library. Classically, molecular biologists have screened for altered protein function *in vivo*, by measuring effects on whole cells. There are, however, several possible advantages to using *in vitro* selections. For example, we no longer require a selection which modifies the growth of a host organism and can instead focus on an isolated function, such as a ligand binding to its receptor. However, *in vitro* selection does require that we connect each member of the protein library to its gene so that we can readily amplify and identify selectants. Bacteriophage (phage) display provides a simple mechanism to link the protein to its DNA.

Phage display typically utilizes bacteriophage M13. This filamentous phage contains single-stranded DNA encased in a protein coat. In contrast to the spherical viruses discussed in Chapter 16, M13 is long (1–2 μm) and narrow (7 nm) and contains five coat proteins, including approximately 2700 copies of the major coat protein gVIIIp (gene VIII protein) and five copies of the infectivity protein gIIIp. In phage display, the gene encoding the peptide

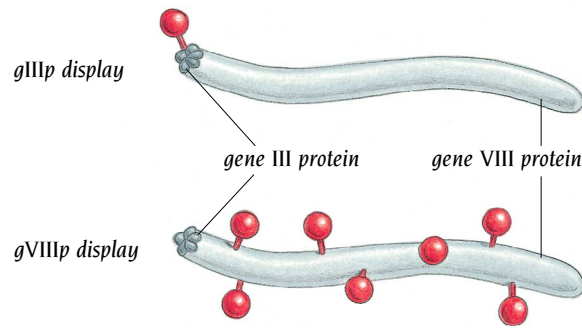


Figure 17.7 Proteins displayed on filamentous phage. In phage display, proteins are usually fused to the major coat protein, gVIIIp (2700 copies per phage), or to the infectivity protein, gIIIp (5 copies per phage). During assembly of the virus in bacteria, capsid fusion proteins are incorporated into the virus and are displayed on the surface of the phage. When gVIIIp fusions are produced, there can be many copies of protein per phage particle, leading to multivalent display. By contrast, gIIIp fusions typically give only one copy per phage (monovalent display).

or protein of interest is usually fused to one of these genes (Figure 17.7). When phage particles are produced in bacterial cells, the capsid fusion protein is incorporated into the viral particle and the phage DNA, containing the gene fusion, is packaged into the phage. The protein phenotype on the phage surface is thereby linked to the DNA genotype within the virus. Since progeny phage will not usually assemble from only capsid fusion proteins, wild-type capsid proteins are also expressed in the bacteria using a so-called helper phage that specifies wild-type capsid proteins but which is deficient in phage packaging. Several copies of the gVIIIp fusion protein molecules can be incorporated into each phage, giving multivalent display, but usually only one gIIIp fusion protein molecule is incorporated, giving monovalent display. A phage-display library results from each bacterium producing phage with a different capsid fusion protein. The typical phage display experiment includes three steps, shown in Figure 17.8. First, a library containing around 10^8 phage is screened for binding to an immobilized target. The selected phage are then propagated in bacteria and the phage DNA is characterized to determine the sequence of the gene corresponding to the mutated binding

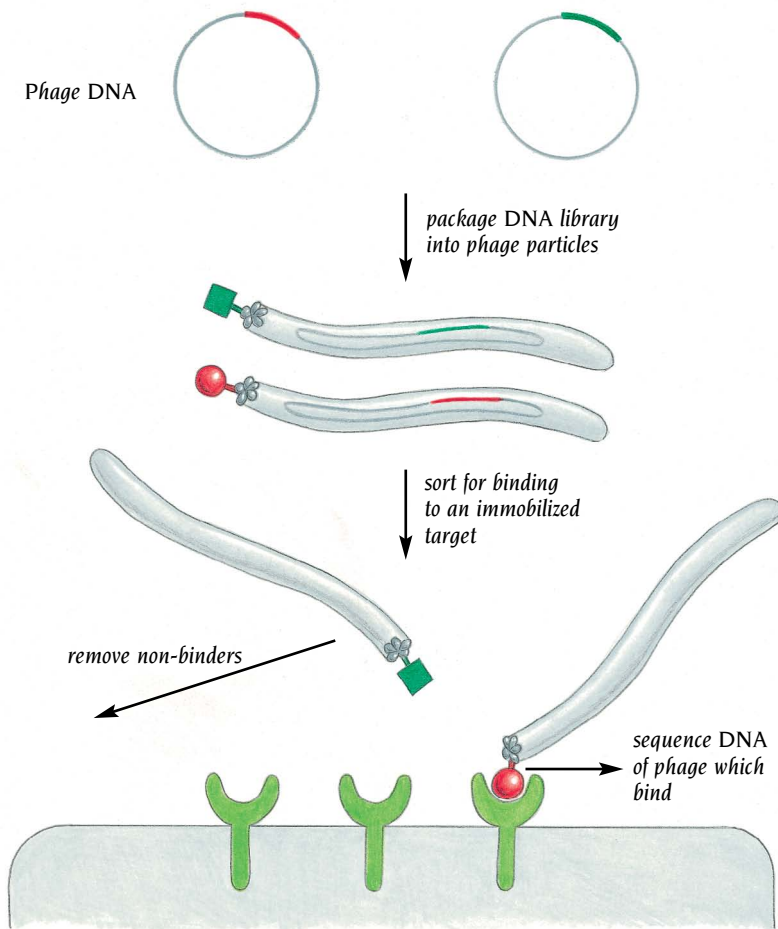


Figure 17.8 Sorting phage display libraries. Each phage particle in a library contains one protein-phage fusion and its corresponding DNA code. This phage library is added to an immobilized target protein; if a fusion protein does not bind the target, the phage displaying that protein is washed away. If a fusion protein does bind, its phage is eluted from the target, propagated in bacteria, and resorted under more stringent binding conditions. After several rounds of this sorting procedure, the DNA from phage are sequenced in order to determine the amino acid sequence of the selected protein.

protein. These three steps can then be repeated after mutagenesis of the selected genes to improve further the properties of the selected proteins.

In 1985, George Smith at the University of Missouri first demonstrated that peptide–phage fusions could be selected through the binding of the peptide to an antibody immobilized on a plate. Since that time, phage display has been used to improve the affinity and specificity of both antibodies and antigens, hormones and receptors; researchers have also studied the interactions of proteins with small molecules or nucleic acids. In the sections below, we focus on the use of phage display to characterize the interactions between proteins, including examples in which a protein scaffold is mutated to change the ligand specificity, a truncated protein is mutated to construct a minimized binding domain, and a random peptide library is sorted to identify a novel receptor agonist.

Affinity and specificity of proteinase inhibitors can be optimized by phage display

The blood coagulation cascade involves several trypsin-like serine proteinases (see Chapter 11) including plasmin, kallikrein, factor XIa, and the tissue factor–factor VIIa (TF–FVIIa) complex. While there would be important clinical benefits to engineering specific protease inhibitors, the active sites of these enzymes are highly conserved, showing as high as 81% identity, making the design of specific inhibitors difficult. **Kunitz domains** make up one family of protein inhibitors of the trypsin-like proteinases (Figure 17.9). These protein domains of approximately 60 residues, stabilized by three disulfide bonds, maintain a highly conserved structure with a sequence identity as low as 33%. Each Kunitz domain recognizes one or more proteinases through a set of 10–14 residues, most of which are in the “binding loop” (residues 11–19, green in Figure 17.9).

As described in Chapter 11, the active site cleft of a serine protease forms a row of subsite pockets, named S5 through S4′, which fit the substrate residues, numbered P5 through P4′. Table 17.1 gives the correlation between subsite and residue number for APPI (Alzheimer’s amyloid β -protein precursor inhibitor) and LACI-D1 (lipoprotein-associated coagulation inhibitor D1) Kunitz domains. The major binding determinant of inhibitors for all trypsin-like proteinases is the presence of Lys 15 or Arg 15 at the P1 site; the specificity of Kunitz domains for different enzymes is then determined by the other subsite residues. Phage display experiments have identified specific Kunitz domain inhibitors with mutations in the subsite residues of the primary binding loop. From the sequences of these specific inhibitors, we can learn the rules governing the recognition between Kunitz domains and serine proteinases.

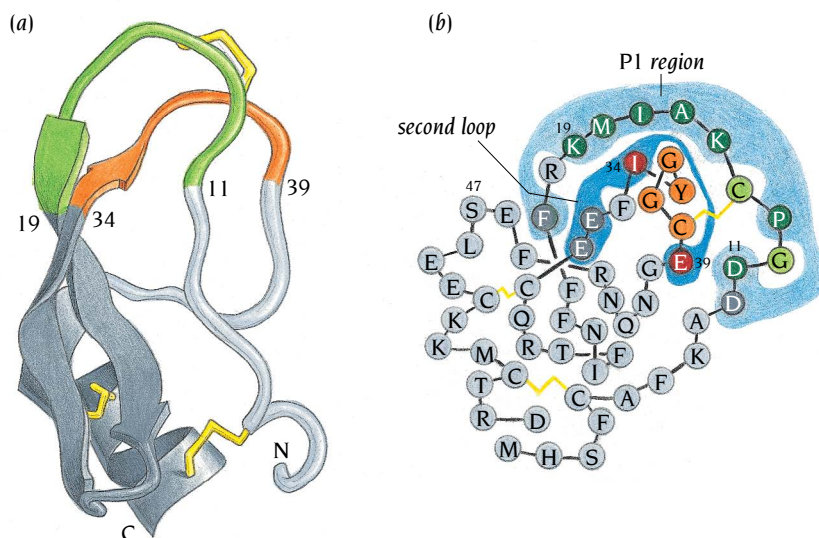


Figure 17.9 Structure and protease-binding properties of Kunitz domains. (a) Structure of APPI determined by x-ray crystallography. This 58 residue protein domain is characteristic of Kunitz domains. The three disulfide bonds are colored yellow. Residues 11–19, colored green, constitute the primary loop involved in binding to trypsin-like serine proteinases. Residues 34–39, colored orange, are also thought to be involved in binding to the proteinase or structuring the binding loop. (b) Amino acid sequence of Kunitz domain LACI-D1. Residues in Kunitz domains making the principal contacts with trypsin-like proteinases are shown in dark colors, colored according to (a). While most important proteinase interactions are with the primary binding loop (11–19), the second loop (34–39) contains some residues which contact either the proteinase or the primary binding loop. [(b) Adapted from W. Markland et al., *Biochemistry* 35: 8045–8057, 1996.]

Table 17.1 Phage-optimized sequences of Kunitz domain libraries

<i>wt sequence and target</i>	<i>primary binding loop</i>								<i>secondary loop</i>		K_i for target (M)
	P ₅	P ₄	P ₃	P ₁	P ₁ '	P ₂ '	P ₃ '	P ₄ '			
<i>residue number</i>	11	12	13	15	16	17	18	19	34	39	
LACI-D1	D	G	P	K	A	I	M	K	I	E	
<i>kallikrein</i>	D	G	P	R	A	A	H	P	S	G	40×10^{-12}
APPI	T	G	P	R	A	M	I	S	F	G	
<i>kallikrein</i>	D	G	H	R	A	A	H	P	Y	G	15×10^{-12}
TF-FVIIa + <i>kallikrein</i>	P	G	P	R	A	L	I	L	F	Y	$\sim 2 \times 10^{-9}$
TF-VIIa	P	G	P	K/R	A	L/M	M	K	I	Y/H	$\sim 10 \times 10^{-9}$

The sequences of LACI-D1 and APPI Kunitz domain protease-binding regions are shown, with the sequences of phage-optimized kallikrein and TF-FVIIa-binding variants given below. Variants of both LACI-D1 and APPI were selected for binding to kallikrein, and additionally APPI variants that bound TF-FVIIa and kallikrein were further selected for TF-VIIa preferential binding.

Two groups have selected phage-displayed Kunitz domains for binding to kallikrein. Mark Dennis and Robert Lazarus at Genentech, US, used the human Kunitz domain APPI as a scaffold. They made three libraries, each containing four or five randomized codons, and combined the selected mutations from all libraries to form a consensus sequence. The consensus sequence, given in Table 17.1, showed an inhibition constant (K_i) for kallikrein of 15 pM, which was lower than the K_i of any of the individual libraries. This result is consistent with the additivity principle which states that the effects of noninteracting mutations tend to be independent. Robert Ladner and coworkers at Protein Engineering Corporation, US, used a different human Kunitz domain, LACI-D1 as a scaffold for their kallikrein-binding libraries. These researchers designed DNA libraries using a restricted set of codons (see Figure 17.6) based on the residues commonly found in Kunitz domains at each position. This strategy reduces diversity, thus permitting more residues to be randomized in each library. However, important interactions might be missed when only a subset of amino acids are available. Nevertheless, the phage selectants identified by this method had very similar sequences and K_i values (40 pM) to the proteins selected at Genentech. The fact that these two phage display libraries arrived at very similar sequences, having started from different scaffolds and library designs, can be described as convergent *in vitro* evolution.

Dennis and Lazarus also sorted their APPI libraries against TF-FVIIa. They found that the tightest inhibitors, (K_i approximately 2 nM), also inhibited factor XIa and kallikrein. To identify inhibitors specific for TF-FVIIa, they employed a competitive, or subtractive, sorting strategy in which soluble competitor (factor XIa) was added during sorting. Kunitz domains that had a high affinity for soluble factor XIa thus remained in solution and were not selected for binding to immobilized TF-FVIIa. Selectants from this competitive sorting maintained nanomolar affinity for TF-FVIIa but inhibited both factor XIa and kallikrein ~1000-fold more weakly. Subtractive sorting thus shifted *in vitro* selection towards mutations that were tolerated only by the desired target.

How do the mutations identified by phage display improve binding specificity? There is as yet no direct structural information on the phage-selected inhibitors; however they can be modeled using data from the crystal structures of other Kunitz domains bound to serine proteinases. These studies lead to the conclusion that the mutations identified by phage display improve binding specificity by maximizing complementarity between the

primary binding loop and the proteinase active site. Analysis of the models of these new Kunitz domains with altered specificity has given novel insights into the mechanisms of Kunitz domain–protease specificity.

Structural scaffolds can be reduced in size while function is retained

In most proteins, only a small number of residues are directly involved in ligand or receptor binding; the rest of the protein provides the three-dimensional structure to position correctly the functional parts of the molecule. James Wells and colleagues at Genentech have used phage display to ask whether a binding epitope can be transferred to a smaller scaffold. This is an interesting question because protein scaffolds often seem larger than they need to be, and also the production of useful proteins is most efficient if the proteins are small. One such minimization involved the Z domain of bacterial protein A, shown in Figure 17.10. The 59-residue Z domain forms an antiparallel three-helix bundle that binds to the Fc portion of IgG (see Chapter 15) with a dissociation constant (K_d) of 20 nM. The Fc-binding epitope is discontinuous, involving residues from both helix 1 and helix 2. The third helix does not contact Fc, but is required to maintain the structure of the binding domain. Minimizing the Z domain thus poses a difficult design problem: a smaller version must maintain the correct three-dimensional placement of functional groups that are not adjacent in sequence. Furthermore, protein structures smaller than 50 residues are relatively rare, and only recently have 30 residue peptides been designed as stable domains (see below).

Andrew Braisted and J.A. Wells prepared phage containing Z domain helices 1 and 2 and restored Fc binding of this 38 residue minidomain in three iterative stages (see Figure 17.10). The truncated peptide was first randomized at four hydrophobic residues which contact helix 3 in the complete Z domain. The consensus sequence from this library maintained the wild-type residues Ile 17 and Leu 23 while the hydrophobic residues Leu 20 and

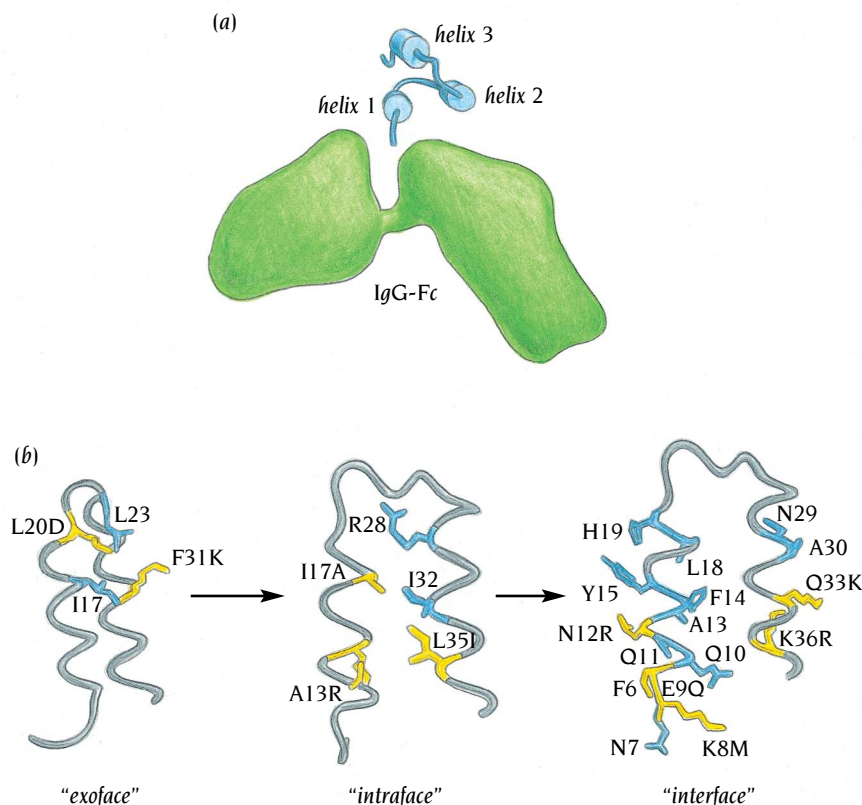


Figure 17.10 Construction of a two helix truncated Z domain. (a) Diagram of the three-helix bundle Z domain of protein A (blue) bound to the Fc fragment of IgG (green). The third helix stabilizes the two Fc-binding helices. (b) Three phage-display libraries of the truncated Z-domain peptide were selected for binding to the Fc. First, four residues at the former helix 3 interface ("exoface") were sorted; the consensus sequence from this library was used as the template for an "intraface" library, in which residues between helices 1 and 2 were randomized. The most active sequence from this library was used as a template for five libraries in which residues on the Fc-binding face ("interface") were randomized. Colored residues were randomized; blue residues were conserved as the wild-type amino acid while yellow residues reached a nonwild-type consensus. [(b) Adapted from A.C. Braisted and J.A. Wells, *Proc. Natl. Acad. Sci. USA* 93: 5688–5692, 1996.]

Phe 31 were mutated to the charged residues Asp and Lys, respectively. This double mutant bound Fc with a K_d of 3.4 μM , a greater than a hundredfold improvement over the unmutated fragment, and had a large increase in α -helical content as shown by circular dichroism spectroscopy. This face of the two-helix peptide was thus converted from a protein core to a protein surface. The second library fixed Asp 20 and Lys 31 and randomized five positions at the “intraface” of helix 1 and helix 2. From this library, three new residues were selected at the open end of the intrahelix interface, and the K_d for this peptide was around 300 nM, a tenfold improvement over the first library. Finally, five libraries were randomized at the Fc-binding face of the two-helix bundle and the consensus mutations were combined. Some libraries yielded mutations that improved binding 2–3 fold, and all together the two-helix bundle contained 12 mutations from the Z domain sequence. A final truncation of the five N-terminal residues yielded a 33 residue peptide with a K_d of 40 nM, very close to the wild-type value of 20 nM. X-ray crystallography and NMR spectroscopy indicated that the two helix bundle had the same three-dimensional structure and the same Fc-binding epitope as the Z domain. Thus, iterative cycles of phage display were used to create a more stable protein surface and to repack the protein core. The resulting scaffold was half the size of the native domain but maintained the three-dimensional arrangement of Fc-binding residues at the interface.

Phage display of random peptide libraries identified agonists of erythropoietin receptor

Many biological processes are activated by hormone–receptor interactions, and a great deal of research has been devoted to identifying peptides or small molecules which either inhibit or simulate hormone function. The idea that a small molecule can mimic the function of a large protein is based on the “hot-spot” principle that a small number of residues at a binding interface contribute most of the binding energy. Nicholas Wrighton and coworkers at Affymax, US, in collaboration with scientists at Johnson & Johnson and at the Scripps Institute, have used phage display methods to isolate a 20 residue peptide, called EMP1, which mimics the activity of erythropoietin (EPO) by promoting dimerization and activation of erythropoietin receptor (EPOR; the extracellular domain of which is called EBP). Erythropoietin is a cytokine hormone which stimulates formation of red blood cells (see Chapter 13).

EMP1 was selected by two cycles of phage display. First, random peptide libraries of the sequence CX_8C , where X is any residue and the cysteines form an intramolecular disulfide bond, were displayed as gVIIIp fusions (see Figure 17.7). Multivalent gVIII protein fusion display permitted selection of weak binders by avidity (multiple) binding to immobilized dimers of EBP. These weakly binding peptides were expanded to the form $\text{X}_5\text{CX}_8\text{CX}_3$ and partially randomized in the X_8 residues. Fusion of the new library to gIIIp yielded a lower valency of display and allowed selection of tight binders. EMP1, isolated from this library, bound to EBP with a K_d of 0.2 μM and stimulated EPOR activity *in vivo*. The dimerization of EMP1, critical for its agonist activity, was probably selected through the interactions of the multivalent peptides with the immobilized EBP dimer.

The crystal structure of EMP1 bound to EBP shows the remarkable structural economy of the EMP1 dimer (Figure 17.11). Each peptide monomer forms a β hairpin, stabilized by an intramolecular disulfide bond. The peptide dimer forms a four-stranded β sheet maintained by four main-chain hydrogen bonds and by the packing of hydrophobic residues. Each monomer makes hydrogen bonds and hydrophobic contacts to both EBP receptors, forming a total of 20 interactions between the peptide dimer and the two EBP proteins; most of the peptide is directly involved in binding (Figure 17.12). Furthermore, EMP1 seems to mimic the binding interactions used by EPO itself. Although there is no structural information on the EPO–EBP complex, the structure of another cytokine hormone, human growth hormone, bound

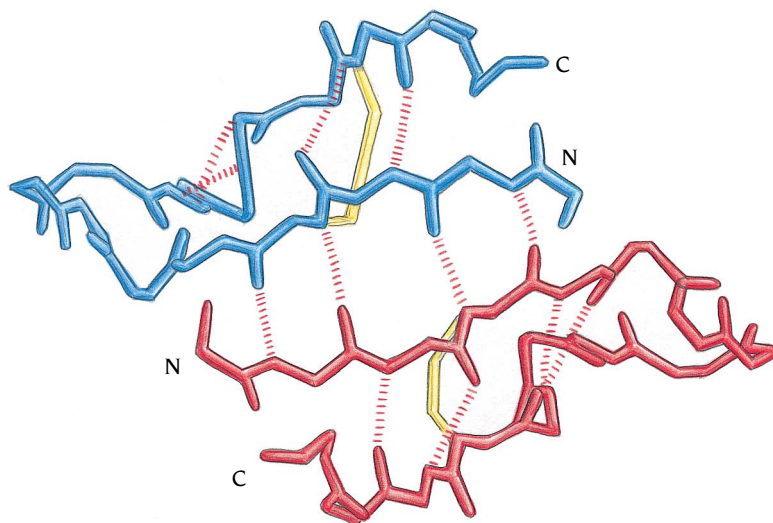


Figure 17.11 Structure of EMP1 dimer from x-ray crystallography. In the presence of EBP, the EMP1 peptide forms a dimer. Each monomer (shown in red and blue) forms a β hairpin structure stabilized by hydrogen bonds (red dashes) and a disulfide bond (yellow). The two peptides form a symmetrical dimer stabilized by four hydrogen bonds (red dashes) and hydrophobic contacts. The two monomers form a four-stranded, anti-parallel pleated sheet.

to its receptor (see Chapter 13) was used as a model. Like EPO, EMP1 contacts four of the six loops on EBP that have sequence similarities to loops of the growth hormone receptor involved in hormone binding. In particular, the three residues of the growth hormone receptor most critical for binding of growth hormone, which correspond to residues Phe 93, Phe 205, and Met 150 in EBP, are well-buried in the EMP1-EBP crystal structure.

EMP1, selected by phage display from random peptide libraries, demonstrates that a dimer of a 20-residue peptide can mimic the function of a monomeric 166-residue protein. In contrast to the minimized Z domain, this selected peptide shares neither the sequence nor the structure of the natural hormone. Thus, there can be a number of ways to solve a molecular recognition problem, and combinatorial methods such as phage display allow us to sort through a multitude of structural scaffolds to discover novel solutions.

DNA shuffling allows accelerated evolution of genes

Natural selection works through the complementary processes of mutation and genetic reassortment by **recombination**. The oligonucleotide-directed mutagenesis methods used in the foregoing examples do not allow for recombination; instead, mutations are combined manually to optimize a protein sequence. Willem Stemmer at Maxygen invented a method of directed evolution that uses both mutation and recombination. This method, called

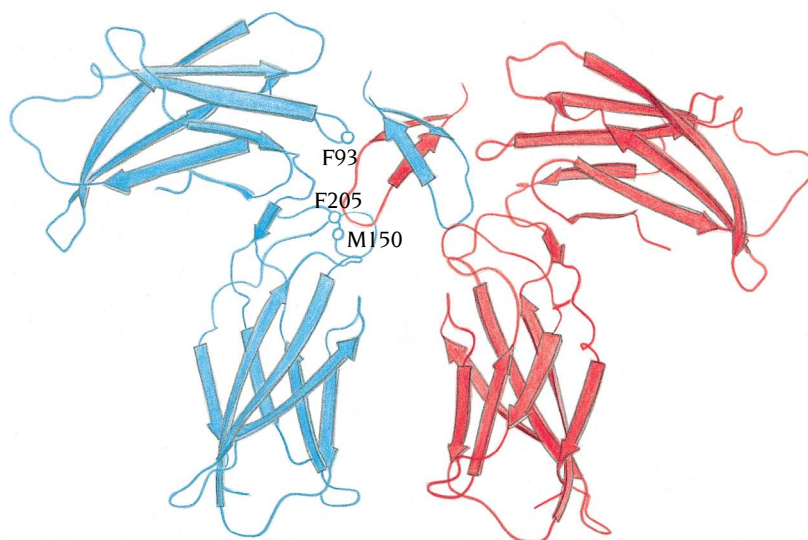
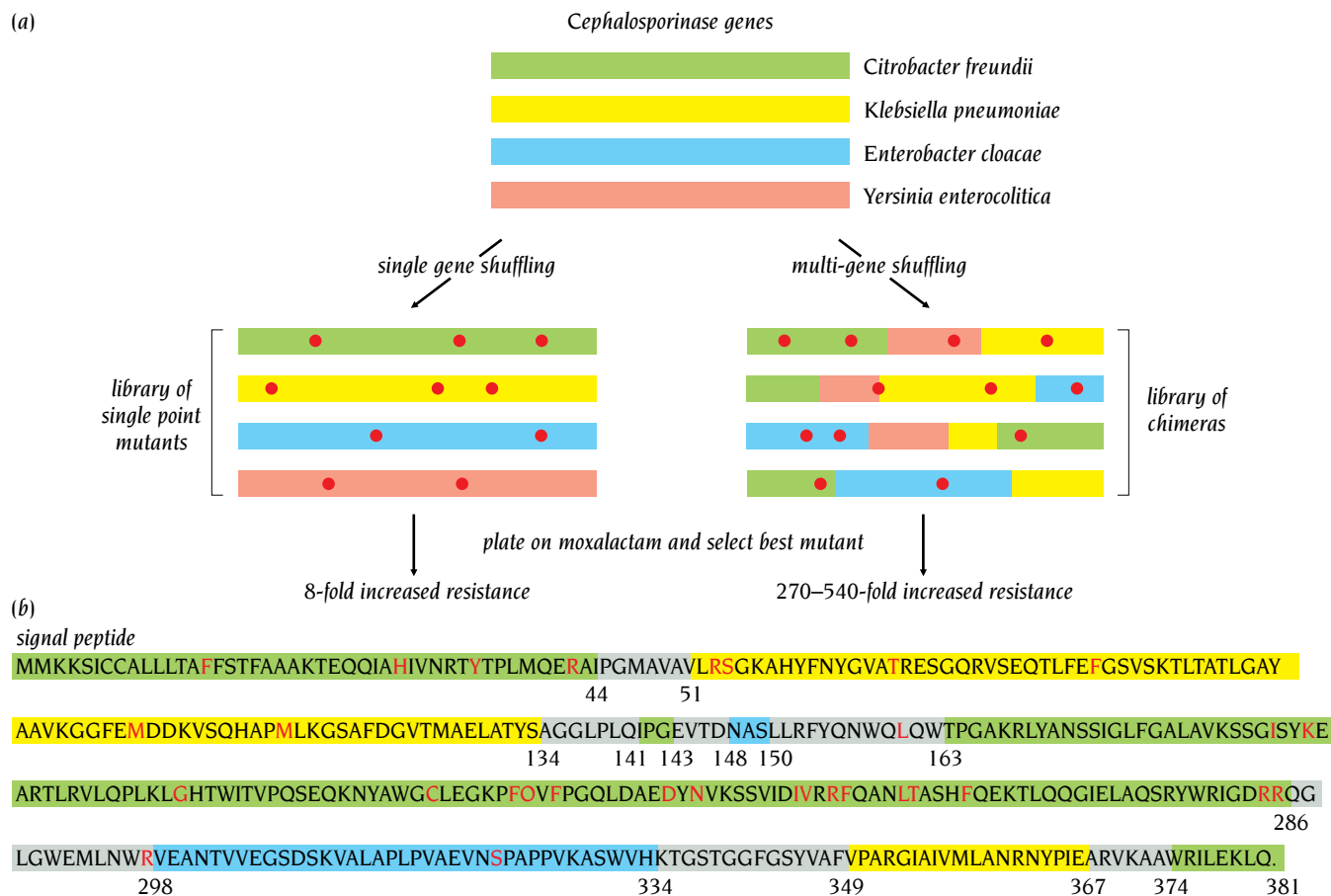


Figure 17.12 Ribbon diagram of EMP1 bound to the extracellular domain of the erythropoietin receptor (EBP). Binding of EMP1 causes dimerization of erythropoietin receptor. The x-ray crystal structure of the EMP1-EBP complex shows a nearly symmetrical dimer complex in which both peptide monomers interact with both copies of EBP. Recognition between the EMP1 peptides and EBP utilizes more than 60% of the EMP1 surface and four of six loops in the erythropoietin-binding pocket of EBP. In particular, three residues thought to be critical for binding erythropoietin (Phe 93, Met 150, and Phe 205) are fully buried in the structure of the peptide-receptor complex. (From J.A. Wells, *Science* 273: 449-450, 1996.)



DNA shuffling (see Figure 17.6), begins by randomly cutting a gene into fragments about 100–300 base pairs in length. This DNA pool is then reassembled with a DNA polymerase, which combines fragments from different copies of the original gene. This process also incorporates point mutations at a defined rate. When point mutations originally in different copies of a gene end up in the same piece of DNA, *in vitro* recombination has occurred. These shuffled DNA libraries are then sorted *in vitro* by phage display or selected *in vivo* through bacterial selection, as described below.

Stemmer and coworkers have extended DNA shuffling to include homologous genes from different organisms. In their first attempt, they mixed the genes encoding class C cephalosporinases from four species (the DNA being 58–82% identical). Their goals were to select for bacterial resistance to the antibiotic moxalactam and to compare the evolution of individual genes with a shuffled gene family. As shown in Figure 17.13, one cycle of evolution using all four genes resulted in recombination of DNA segments as well as incorporation of random point mutations. By contrast, one cycle of DNA shuffling of a single gene only introduced point mutations. The results of recombination of different genes was dramatic: the single genes yielded eightfold increases in antibiotic resistance, while the four genes together gave 270 to 540-fold improvements. The best clone contained eight discrete DNA segments from three of the four genes as well as 33 point mutations. Figure 17.14 shows a three-dimensional model based on the crystal structure of one of the native enzymes. The different colors identify the origin of each protein segment; interestingly, these segments form units of secondary structure. This example demonstrates the utility of recombination and mutation in directed evolution. Additionally, this work underscores the power of combinatorial techniques to construct enzymes whose design would be impossible given our current understanding of the factors governing protein structure and function.

Figure 17.13 DNA shuffling of cephalosporinase genes. (a) Four homologous genes from bacteria were subjected to one cycle of cleavage and recombination, resulting in mutant genes containing point mutations (shown in red circles) and large pieces of each wild-type sequence. These mutant genes were transformed into bacteria and screened for resistance to the antibiotic moxalactam; the most active mutants increased resistance by 540-fold over the wild-type genes. By contrast, DNA shuffling of each individual gene yielded only point mutations; the most active of these mutants only increased resistance to moxalactam eightfold. (b) Amino acid sequence of the most active mutant, colored as in (a) to show the origin of regions of the protein, with point mutations shown in red. The gray regions cannot be unambiguously assigned to one of the original cephalosporinase genes. (From A. Cramer et al., *Nature* 391: 288–291, 1998.)



Figure 17.14 Model of evolved mutant from cephalosporinase shuffling. The sequence of the most active cephalosporinase mutant was modeled using the crystal structure of the class C cephalosporinase from *Enterobacter cloacae*. The mutant and wild-type proteins were 63% identical. This chimeric protein contained portions from three of the starting genes, including *Enterobacter* (blue), *Klebsiella* (yellow), and *Citrobacter* (green), as well as 33 point mutations (red). (Courtesy of A. Cramer.)

Protein structures can be designed from first principles

The ultimate goal of protein engineering is to design an amino acid sequence that will fold into a protein with a predetermined structure and function. Paradoxically, this goal may be easier to achieve than its inverse, the solution of the folding problem. It seems to be simpler to start with a three-dimensional structure and find one of the numerous amino acid sequences that will fold into that structure than to start from an amino acid sequence and predict its three-dimensional structure. We will illustrate this by the design of a stable zinc finger domain that does not require stabilization by zinc.

The classic zinc fingers, the DNA-binding properties of which are discussed in Chapter 10, are small compact domains of about 30 residues that fold into an antiparallel β hairpin followed by an α helix. All known classic zinc fingers have a zinc atom bound to two cysteines in the hairpin and two histidines in the helix, creating a sequence motif common to all zinc finger genes. In the absence of zinc the structure is unfolded.

Stephen Mayo and Bassil Dahiyat at the California Institute of Technology asked the question: Is it possible to design from first principles a sequence whose main chain obtains this zinc finger fold without a zinc atom to stabilize the structure? They chose the second zinc finger of Zif 268 (see Chapter 10) with 28 residues as their target fold and applied their recently developed computer algorithm to the problem. Briefly, this algorithm searches through a very large number of possible sequences using a fast selection procedure for those sequences that stabilize a given fold.

On the basis of the template fold, side chain positions are divided into three categories: core, surface, and boundary positions. Allowed residues at the core positions are Ala, Val, Leu, Ile, Phe, Tyr and Trp, and at the surface positions Ala, Ser, Thr, His, Asp, Asn, Glu, Gln, Lys and Arg. The combined

Table 17.2 Amino acid sequences of the second zinc finger of Zif 268 and the designed peptide FSD-1

	1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
--	---	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

core and surface sets (16 residues since Ala belongs to both) are allowed at the boundary positions. They are thus using a restricted set of residues where Pro, Cys, and Met are absent and the only hydrophobic residue allowed at the surface is Ala. Gly is used in special positions to minimize backbone strain due to the conformation of the template.

The selection algorithm used involves energy functions for van der Waals interactions, hydrogen bonding and solvation combined with a secondary structure propensity potential. The total number of amino acid sequences that must be screened is the product of the number of possible amino acids at each residue position. For Zif 268 there is one core position with seven possible amino acids, 20 surface positions each with 10 possible amino acids and seven boundary positions each with 16 possible amino acids, giving a total of about 10^{27} possible sequences. This virtual library size is 15 orders of magnitude larger than an experimental combinatorial library using current methods. A corresponding peptide library consisting of only a single molecule for each 28-residue sequence would have a mass of 11,600 kilograms. The actual virtual search space is, however, even larger since different conformations, rotamers, must be considered for every possible side chain.

The optimal sequence obtained, called FSD-1 for full sequence design, is shown in Table 17.2 and compared with the sequence of the template Zif 268. A search of the FSD-1 sequence against protein databases did not reveal a statistically significant similarity with any other protein, including zinc finger proteins.

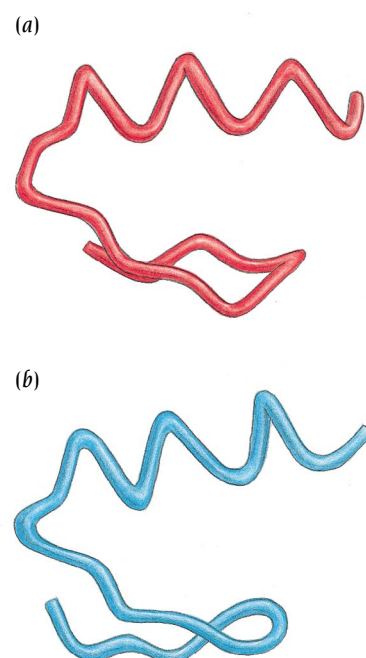
In order to examine whether this sequence gave a fold similar to the template, the corresponding peptide was synthesized and its structure experimentally determined by NMR methods. The result is shown in Figure 17.15 and compared to the design target whose main chain conformation is identical to that of the Zif 268 template. The folds are remarkably similar even though there are some differences in the loop region between the two β strands. The core of the molecule, which comprises seven hydrophobic side chains, is well-ordered whereas the termini are disordered. The root mean square deviation of the main chain atoms are 2.0 Å for residues 3 to 26 and 1.0 Å for residues 8 to 26.

In addition to being a remarkable demonstration of the power of computer-based combinatorial design of a protein fold, this designed peptide is the shortest known peptide consisting entirely of naturally occurring amino acids that folds into a well-ordered structure without metal binding, oligomerization or disulfide bond formation.

A β structure has been converted to an α structure by changing only half of the sequence

In 1994 a prize of \$1000 called the Paracelsus challenge was offered to any designer of protein structures who could convert one protein fold into another while retaining 50% of the original sequence. The spirit of this challenge was to take one important step towards solving the folding problem by assessing the fraction of a protein's amino acid sequence that is sufficient to specify its structure. On the basis of our knowledge of naturally occurring proteins, this challenge did not look like a feasible project: all proteins of known structure with more than 30% sequence identity had been shown to have identical folds. Yet, it did not take more than three years before the prize

Figure 17.15 Schematic diagrams of the main-chain conformations of the second zinc finger domain of Zif 268 (red) and the designed peptide FSD-1 (blue). The zinc finger domain is stabilized by a zinc atom whereas FSD-1 is stabilized by hydrophobic interactions between the β strands and the α helix. (Adapted from B.I. Dahiyat and S.L. Mayo, *Science* 278: 82–87, 1997.)



with high preference for β sheet formation were replaced with residues with high preference for α helix formation, such as Tyr to Lys and Leu to Ala, in regions that were required to switch conformation. Since Rop is a coiled-coil protein, hydrophobic residues were incorporated in appropriate *a* and *d* positions of the heptad repeat (see Chapter 3). Among other changes was the introduction of an intramonomer salt bridge between Arg 16 and Asp 46 that is present between the two helices of Rop.

During this process of designing sequence changes, models were built and assessed to ensure that there were no obvious steric clashes and that the hydrophobic core was well packed. Furthermore, secondary structure prediction was also used to monitor the progress of change and to choose among different possible substitutions. The final sequence (see Table 17.3) contains 28 changes; it had 50% identity to B1 and the similarity to Rop had increased from 5.4% identity to 41%.

A gene encoding this sequence was synthesized and the corresponding protein, called Janus, was expressed, purified, and characterized. The atomic structure of this protein has not been determined at the time of writing but circular dichroic and NMR spectra show very clear differences from B1 and equally clear similarities to Rop. The protein is a dimer in solution like Rop and thermodynamic data indicate that it is a stably folded protein and not a molten globule fold like several other designed proteins.

These results indicate that it is possible to change the fold of a protein by changing a restricted set of residues. They also confirm the validity of the rules for stability of helical folds that have been obtained by analysis of experimentally determined protein structures. One obvious implication of this work is that it might be possible, by just changing a few residues in Janus, to design a mutant that flip-flops between α helical and β sheet structures. Such a polypeptide would be a very interesting model system for prions and other amyloid proteins.

Conclusion

Homologous proteins have similar three-dimensional structures. They contain a core region, a scaffold of secondary structure elements, where the folds of the polypeptide chains are very similar. Loop regions that connect the building blocks of the scaffolds can vary considerably both in length and in structure. From a database of known immunoglobulin structures it has, nevertheless, been possible to predict successfully the conformation of hyper-variable loop regions of antibodies of known amino acid sequence.

Methods for the prediction of the secondary structure of a set of homologous proteins can reach an accuracy of about 75%, most of the errors occur at the ends of α helices or β strands. The central regions of these secondary structure elements are often correctly predicted but the methods do not always correctly distinguish between α helices and β strands.

Prediction of tertiary structure from the amino acid sequence is the major unsolved problem in structural molecular biology. The inverse problem, to predict which amino acid sequences can have a given fold seem to be easier to solve. Significant progress has been made in recent years in threading methods, which assign a known fold to a given sequence by threading the sequence through all known folds.

Protein engineering is now routinely used to modify protein molecules either via site-directed mutagenesis or by combinatorial methods. Factors that are important for the stability of proteins have been studied, such as stabilization of α helices and reducing the number of conformations in the unfolded state. Combinatorial methods produce a large number of random mutants from which those with the desired properties are selected *in vitro* using phage display. Specific enzyme inhibitors, increased enzymatic activity and agonists of receptor molecules are examples of successful use of this method.

Small protein molecules with a predetermined fold can be designed *in silico* by energy calculations of all possible combinations of a restricted set of amino acid residues. A designed zinc finger fold that is stable in the absence of zinc showed no significant sequence similarity to any known protein sequence. Important progress has been made in assessing the fraction of a protein's amino acid sequence that is sufficient to specify its structure. A protein that folds into a mainly β -sheet structure was converted into an α -helical structure by changing only 50% of its sequence.

Selected readings

General

- Alber, T. Mutational effects on protein stability. *Annu. Rev. Biochem.* 58: 765–798, 1989.
- Barton, G.J. Protein secondary structure prediction. *Curr. Opin. Struct. Biol.* 5: 372–376, 1995.
- Blundell, T.L., et al. Knowledge-based prediction of protein structures and the design of novel molecules. *Nature* 326: 347–352, 1987.
- Bowie, J.U., Eisenberg, D. Inverted protein structure prediction. *Curr. Opin. Struct. Biol.* 3: 437–444, 1993.
- DeGrado, W.F., Wasserman, Z.R., Lear, J.D. Protein design, a minimalist approach. *Science* 243: 622–628, 1989.
- Fasman, G.D. Protein conformational prediction. *Trends Biochem. Sci.* 14: 295–299, 1989.
- Fersht, A.R. Protein engineering. *Protein Eng.* 1: 7–16, 1986.
- Fersht, A.R. The hydrogen bond in molecular recognition. *Trends Biochem. Sci.* 12: 301–304, 1987.
- Finkelstein, A.V. Protein structure: what is possible to predict now? *Curr. Opin. Struct. Biol.* 7: 60–71, 1997.
- Jones, D.T., Thornton, J. Potential energy functions for threading. *Curr. Opin. Struct. Biol.* 6: 210–216, 1996.
- Kay, B.K., Winter, J., McCafferty, J. eds. *Phage Display of Peptides and Proteins: a Laboratory Manual*. San Diego: Academic Press, 1996.
- Moult, J., et al. A large-scale experiment to assess protein–structure prediction methods. *Proteins* 23: ii–iv, 1995.
- Murzin, A., et al. Scop: a structural classification of proteins. Database for the investigation of sequences and structures. *J. Mol. Biol.* 247: 536–540, 1995.
- Orengo, C.A., et al. CATH—a hierarchic classification of protein domain structures. *Structure* 5: 1093–1108, 1997.
- Richardson, J.S., Richardson, D.C. The *de novo* design of protein structures. *Trends Biochem. Sci.* 14: 304–309, 1989.
- Sippl, M.J. Knowledge-based potentials for proteins. *Curr. Opin. Struct. Biol.* 5: 229–235, 1995.
- Sonnhammer, E.L., Eddy, S.R., Durbin, R. Pfam: a comprehensive database of protein domain families based on seed alignments. *Proteins* 28: 405–420, 1997.
- Thornton, J.M., Gardner, S.P. Protein motifs and data-base searching. *Trends Biochem. Sci.* 14: 300–304, 1989.
- von Heijne, G. *Sequence Analysis in Molecular Biology: Treasure Trove or Trivial Pursuit*. San Diego: Academic Press, 1987.
- Wells, J.A. Additivity of mutational effects in proteins. *Biochemistry* 29: 8509–8517, 1990.

Specific structures

- Aszodi, A., Taylor, W.R. Homology modelling by distance geometry. *Fold. Des.* 1: 325–334, 1996.
- Bowie, J.U. Deciphering the message in protein sequences: tolerance to amino acid substitutions. *Science* 247: 1306–1310, 1990.
- Bowie, J.U., Luthy, R., Eisenberg, D. A method to identify protein sequences that fold into a known three-dimensional structure. *Science* 253: 164–170, 1991.
- Braisted, A.C., Wells, J.A. Minimizing a binding domain from protein A. *Proc. Natl. Acad. Sci. USA* 93: 5688–5692, 1996.
- Bryant, S.H. Evaluation of threading specificity and accuracy. *Proteins* 26: 172–185, 1996.
- Chothia, C., et al. Conformations of immunoglobulin hypervariable regions. *Nature* 342: 877–883, 1989.
- Chothia, C., et al. The predicted structure of immunoglobulin D 1.3 and its comparison with the crystal structure. *Science* 233: 755–758, 1986.
- Chothia, C., Lesk, A. The relation between the divergence of sequence and structure in proteins. *EMBO J.* 5: 823–826, 1986.
- Chou, P.Y., Fasman, G.D. Prediction of protein conformation. *Biochemistry* 13: 222–245, 1974.

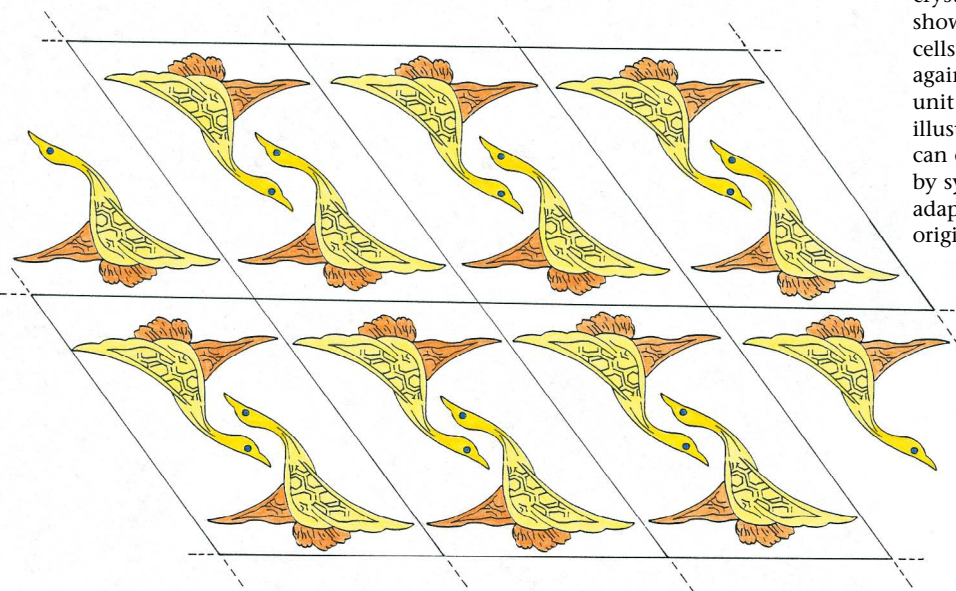
- Cohen, C., Parry, D.A.D. α -helical coiled coils and bundles: how to design an α -helical protein. *Proteins: Struct. Funct. Gen.* 7: 1–15, 1990.
- Cramer, A., Raillard, S.-A., Bermudez, E., Stemmer, W.P.C. DNA shuffling of a family of genes from diverse species accelerates directed evolution. *Nature* 391: 288–291, 1998.
- Dahiyat, B.I., Mayo, S.L. De novo protein design: fully automated sequence selection. *Science* 278: 82–87, 1997.
- Dalal, S., Balasubramanian, S., Regan, L. Protein alchemy: changing β sheet into α helix. *Nature Struct. Biol.* 4: 548–552, 1997.
- DeGrado, W.F., Regan, L., Ho, S.P. The design of a four-helix bundle protein. *Cold Spring Harbor Symp. Quant. Biol.* 52: 521–526, 1987.
- Dennis, M.S., Herzka, A., Lazarus, R.A. Potent and selective Kunitz domain inhibitors of plasma kallikrein designed by phage display. *J. Biol. Chem.* 270: 25411–25417, 1995.
- Dennis, M.S., Lazarus, R.A. Kunitz domain inhibitors of tissue factor-factor VIIa. I. Potent inhibitors selected from libraries by phage display. *J. Biol. Chem.* 269: 22129–22136, 1994.
- Dennis, M.S., Lazarus, R.A. Kunitz domain inhibitors of tissue factor-factor VIIa. II. Potent and specific inhibitors by competitive phage selection. *J. Biol. Chem.* 269: 22137–22144, 1994.
- Faber, H.R., Matthews, B.W. A mutant T4 lysozyme displays five different crystal conformations. *Nature* 348: 263–266, 1990.
- Fersht, A.R., et al. Hydrogen bonding and biological specificity analyzed by protein engineering. *Nature* 314: 235–238, 1985.
- Garnier, J., Osguthorpe, D.J., Robson, B. Analysis of the accuracy and implications of simple methods for predicting the secondary structure of globular proteins. *J. Mol. Biol.* 120: 97–120, 1978.
- Gribskov, M., McLaschlan, A.D., Eisenberg, D.E. Profile analysis: detection of distantly related proteins. *Proc. Natl. Acad. Sci. USA* 84: 4355–4358, 1987.
- Jones, D.T., Taylor, W.R., Thornton, J.M. A new approach to protein fold recognition. *Nature* 358: 86–89, 1992.
- Karpusas, M., et al. Hydrophobic packing in T4 lysozyme probed by cavity-filling mutants. *Proc. Natl. Acad. Sci. USA* 86: 8237–8241, 1989.
- Kellis, J.T., et al. Contribution of hydrophobic interactions to protein stability. *Nature* 333: 784–786, 1988.
- Lim, V.I. Algorithms for prediction of α -helical and β -structural regions in globular proteins. *J. Mol. Biol.* 88: 873–894, 1974.
- Livnah, O., Stura, E.A., Johnson, D.L., Middleton, S.A., Mulcahy, L.S., Wrighton, N.C., Dower, W.J., Jolliffe, L.K., Wilson, I.A. Functional mimicry of a protein hormone by a peptide agonist: the EPO receptor complex at 2.8 Å. *Science* 273: 464–471, 1996.
- Matsumura, M., Signor, G., Matthews, B.W. Substantial increase of protein stability by multiple disulfide bonds. *Nature* 342: 291–293, 1989.
- Matthews, B.W., Nicholson, H., Becktel, W.J. Enhanced protein thermostability from site-directed mutations that decrease the entropy of unfolding. *Proc. Natl. Acad. Sci. USA* 84: 6663–6667, 1987.
- Nicholson, H., Becktel, W.J., Matthews, B.W. Enhanced protein thermostability from designed mutations that interact with α -helix dipoles. *Nature* 336: 651–656, 1988.
- Regan, L., DeGrado, W.F. Characterization of a helical protein designed from first principles. *Science* 241: 976–978, 1988.
- Rice, D.W., et al. A 3D-1D substitution matrix for protein fold recognition that includes predicted secondary structure of the sequence. *J. Mol. Biol.* 267: 1026–1038, 1997.
- Rice, D.W., et al. Fold assignments for amino acid sequences of the CASP2 experiment. *Proteins (Suppl. 1)* 113–122, 1997.
- Richardson, J.S., Richardson, D.C. Amino acid preferences for specific locations at the ends of α helices. *Science* 240: 1648–1652, 1988.
- Rost, B., Sander, C. Combining evolutionary information and neural networks to predict secondary structure. *Proteins* 19: 55–72, 1994.
- Rost, B., Sander, C., Schneider, R. Redefining the goals of protein secondary structure prediction. *J. Mol. Biol.* 235: 13–26, 1994.
- Smith, G.P. Filamentous fusion phage: novel expression vectors that display cloned antigens on the virion surface. *Science* 228: 1315–1317, 1985.
- Starovasnik, M.A., Braisted, A.C., Wells, J.A. Structural mimicry of a native protein by a minimized binding domain. *J. A. Proc. Natl. Acad. Sci. USA* 94: 10080–10085, 1997.
- Stemmer, W.P.C. DNA shuffling by random fragmentation and reassembly: *In vitro* recombination for molecular evolution. *Proc. Natl. Acad. Sci. USA* 91: 10747–10751, 1994.
- Weaver, L.H., Matthews, B.W. Structure of bacteriophage T4 lysozyme refined at 1.7 Å resolution. *J. Mol. Biol.* 193: 189–199, 1987.
- Wells, T.N.C., Fersht, A.R. Hydrogen bonding in enzymatic catalysis analyzed by protein engineering. *Nature* 316: 656–657, 1985.
- Wetzel, R. Harnessing disulfide bonds using protein engineering. *Trends Biochem. Sci.* 12: 478–482, 1987.
- Wrighton, N.C., et al. Small peptides as potent mimetics of the protein hormone erythropoietin. *Science* 273: 458–463, 1996.
- Zvelebil, M.J., et al. Prediction of protein secondary structure and active sites using the alignment of homologous sequences. *J. Mol. Biol.* 195: 957–961, 1987.

The structures described in this book have been determined by physical methods: most of them by x-ray crystallography, some of the smaller ones by nuclear magnetic resonance (NMR). We conclude the book with a short description of these techniques. It is not our aim to convert biologists into x-ray crystallographers and NMR spectroscopists; a complete explanation of the physical basis of these techniques and of the methods as currently practiced would fill more than one textbook. Our purpose is rather to convey the essence of the principles and procedures involved, so as to provide a general understanding of what is entailed in solving protein structures by these means. We will see how deriving a three-dimensional protein structure from x-ray or NMR data depends not only on the quality of the data themselves, but also on biochemical and sometimes genetic information that are essential to their interpretation.

Several different techniques are used to study the structure of protein molecules

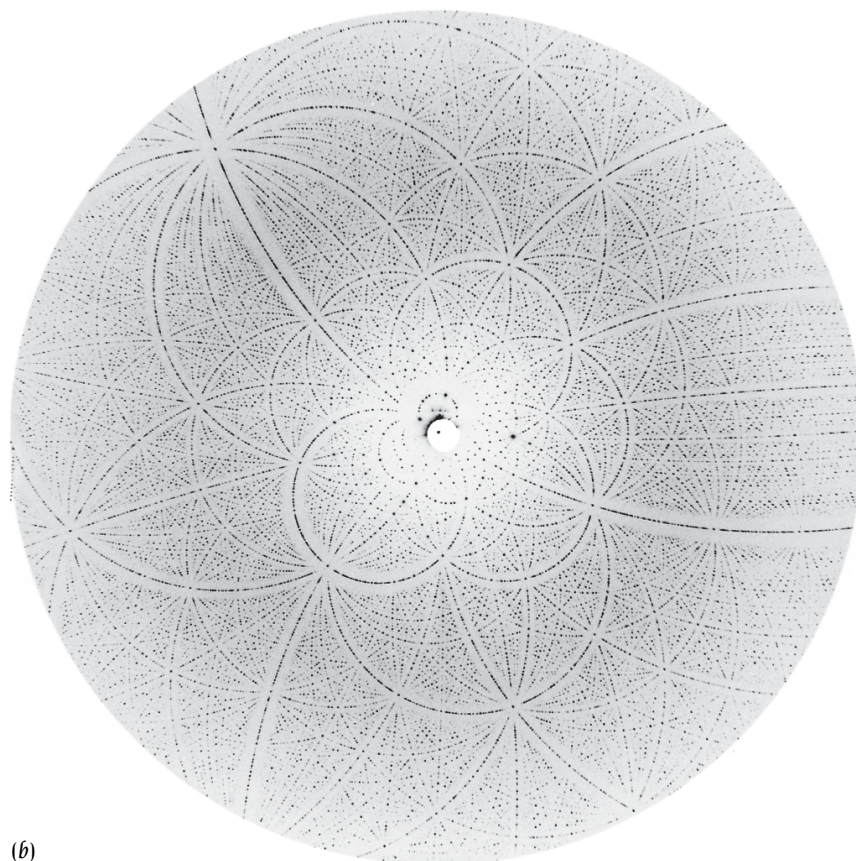
Different techniques give different and complementary information about protein structure. The primary structure is obtained by biochemical methods, either by direct determination of the amino acid sequence from the protein or indirectly, but more rapidly, from the nucleotide sequence of the

Figure 18.1 A crystal is built up from many billions of small identical units, or unit cells. These unit cells are packed against each other in three dimensions much as identical boxes are packed and stored in a warehouse. The unit cell may contain one or more than one molecule. Although the number of molecules per unit cell is always the same for all the unit cells of a single crystal, it may vary between different crystal forms of the same protein. The diagram shows in two dimensions several identical unit cells, each containing two objects packed against each other. The two objects within each unit cell are related by twofold symmetry to illustrate that each unit cell in a protein crystal can contain several molecules that are related by symmetry to each other. (The pattern is adapted from a Japanese stencil of unknown origin from the nineteenth century.)





(a)



(b)

corresponding gene or cDNA. The quaternary structure of large proteins or aggregates such as virus particles, ribosomes, or gap junctions can be determined by **electron microscopy**. In general, this method gives structural information at very low resolution, with no atomic details, although, if one can obtain ordered two-dimensional arrays of the object, the noise in the electron microscopic image can be reduced enough to reveal the shape of individual subunits or, in rare cases, even determine the path of the polypeptide chain within a protein molecule.

To obtain the secondary and tertiary structure, which requires detailed information about the arrangement of atoms within a protein, the main method so far has been x-ray crystallography. In recent years NMR methods have been developed to obtain three-dimensional models of small protein molecules, and NMR is becoming increasingly useful as it is further developed.

Protein crystals are difficult to grow

The first prerequisite for solving the three-dimensional structure of a protein by **x-ray crystallography** is a well-ordered crystal that will diffract x-rays strongly. The crystallographic method depends, as we will see, upon directing a beam of x-rays onto a regular, repeating array of many identical molecules (Figure 18.1) so that the x-rays are diffracted from it in a pattern, a **diffraction pattern**, from which the structure of an individual molecule can be retrieved. The repeating unit forming the crystal is called the **unit cell**, and each unit cell may contain one or more molecules. Well-ordered crystals (Figure 18.2) are difficult to grow because globular protein molecules are large, spherical, or ellipsoidal objects with irregular surfaces, and it is impossible to pack them into a crystal without forming large holes or channels between the individual molecules. These channels, which usually occupy

Figure 18.2 Well-ordered protein crystals diffract x-rays and produce diffraction patterns that can be recorded on film. The crystal shown in (a) is of the enzyme RuBisCo from spinach and the photograph in (b) is a recording (Laue photograph) of the diffraction pattern of a similar crystal of the same enzyme. The diffraction pattern was obtained using polychromatic radiation from a synchrotron source in the wavelength region 0.5 to 2.0 Å. More than 100,000 diffracted beams have been recorded on this film during an exposure of the crystal to x-rays for less than one second. (The Laue photograph was recorded by Janos Hajdu, Oxford, and Inger Andersson, Uppsala, at the synchrotron radiation source in Daresbury, England.)

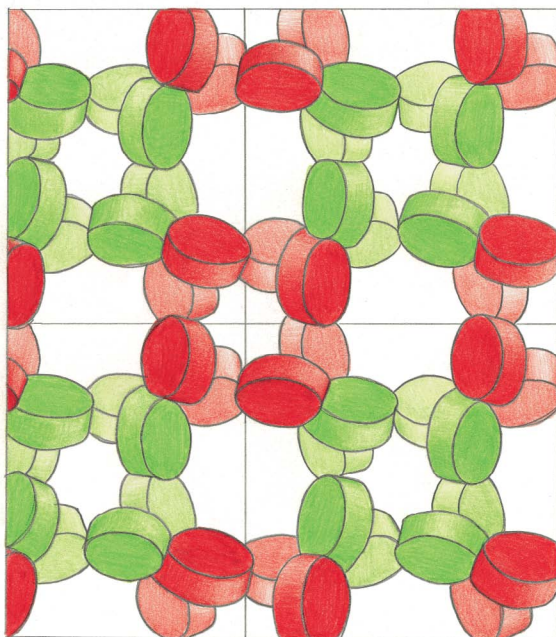


Figure 18.3 Protein crystals contain large channels and holes filled with solvent molecules, as shown in this diagram of the molecular packing in crystals of the enzyme glycolate oxidase. The subunits (colored disks) form octamers of molecular weight around 300 kDa, with a hole in the middle of each of about 15 Å diameter. Between the molecules there are channels (white) of around 70 Å diameter through the crystal. (Courtesy of Ylva Lindqvist, who determined the structure of this enzyme to 2.0 Å resolution in the laboratory of Carl Branden, Uppsala.)

more than half the volume of the crystal, are filled with disordered solvent molecules (Figure 18.3). The protein molecules are in contact with each other at only a few small regions, and even in these regions many interactions are indirect, through one or several layers of solvent molecules, usually water. This is one reason why structures of proteins determined by x-ray crystallography are the same as those for the proteins in solution.

Crystallization is usually quite difficult to achieve, and crystal growth can be slow; in some cases it may require months for sufficiently large crystals (~0.5 mm) to grow from microcrystals. The formation of crystals is also critically dependent on a number of different parameters, including pH, temperature, protein concentration, the nature of the solvent and precipitant as well as the presence of added ions or ligands to the protein. Many crystallization experiments are therefore required to screen all these parameters for the few combinations that might give crystals suitable for x-ray diffraction analysis. Crystallization robots and commercially available crystallization kits automate and speed up the tedious work of reproducibly setting up large numbers of crystallization experiments.

A pure and homogeneous protein sample is crucial for successful crystallization, and recombinant DNA techniques have been a major breakthrough in this regard. Proteins obtained from cloned genes in efficient expression vectors can be purified quickly to homogeneity in large quantities in a few purification steps. As a rule of thumb, a protein to be crystallized should ideally be more than 97% pure according to standard criteria of homogeneity. Crystals form when molecules are precipitated very slowly from **supersaturated solutions**. The most frequently used procedure for making protein crystals is the **hanging-drop** method (Figure 18.4), in which a drop of protein solution is brought very gradually to supersaturation by loss of water from the droplet to the larger reservoir that contains salt or polyethylene glycol solution.

Since there are so few direct packing interactions between protein molecules in a crystal, small changes in, for example, the pH of the solution can cause the molecules to pack in different ways to produce different crystal forms. The structures of some protein molecules such as lysozyme and myoglobin have been determined in different crystal forms and found to be essentially similar, except for a few side chains involved in packing interactions. Because they are so few, these interactions between protein molecules in a crystal do not change the overall structure of the protein. However,

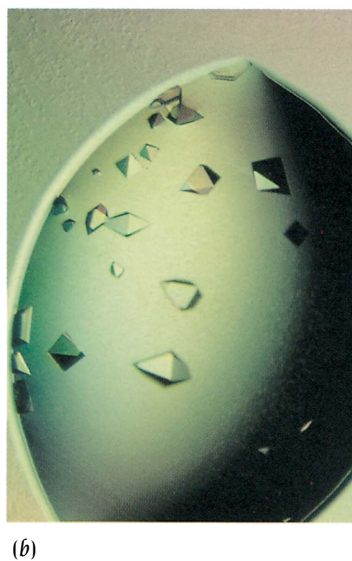
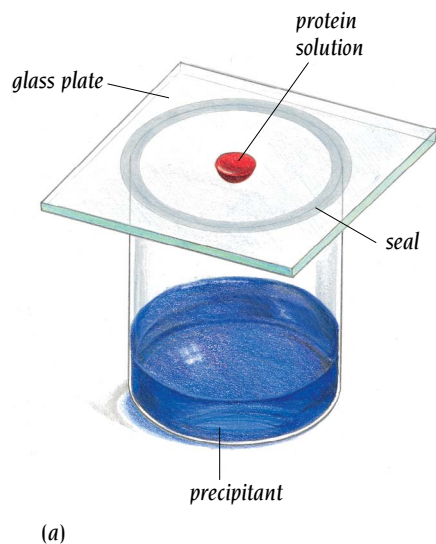


Figure 18.4 The hanging-drop method of protein crystallization. (a) About 10 μl of a 10 mg/ml protein solution in a buffer with added precipitant—such as ammonium sulfate, at a concentration below that at which it causes the protein to precipitate—is put on a thin glass plate that is sealed upside down on the top of a small container. In the container there is about 1 ml of concentrated precipitant solution. Equilibrium between the drop and the container is slowly reached through vapor diffusion, the precipitant concentration in the drop is increased by loss of water to the reservoir, and once the saturation point is reached the protein slowly comes out of solution. If other conditions such as pH and temperature are right, protein crystals will occur in the drop. (b) Crystals of recombinant enzyme RuBisCo from *Anacystis nidulans* formed by the hanging-drop method. (Courtesy of Janet Newman, Uppsala, who produced these crystals.)

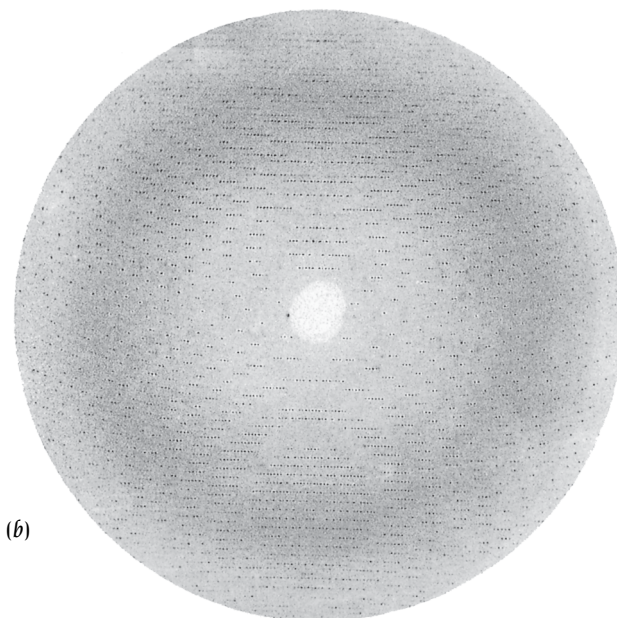
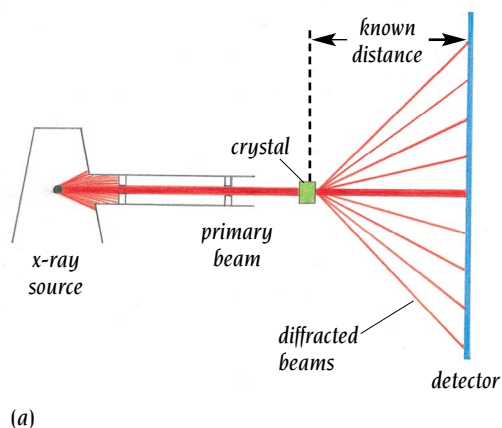
different crystal forms can be more or less well ordered and hence give diffraction patterns of different quality. As a general rule, the more closely the protein molecules pack, and consequently the less water the crystals contain, the better is the diffraction pattern because the molecules are better ordered in the crystal.

X-ray sources are either monochromatic or polychromatic

X-rays are electromagnetic radiation at short wavelengths, emitted when electrons jump from a higher to a lower energy state. X-rays can be produced by high-voltage tubes in which a metal plate, the anode, is bombarded with accelerating electrons and thereby caused to emit x-rays of a specific wavelength, so-called **monochromatic x-rays**. The high voltage rapidly heats up the metal plate, which therefore has to be cooled. Efficient cooling is achieved by so-called rotating anode x-ray generators, where the metal plate revolves during the experiment so that different parts are heated up. Rotating anode x-ray generators are the conventional equipment used in most protein crystallography laboratories.

More powerful x-ray beams can be produced in synchrotron storage rings where electrons (or positrons) travel close to the speed of light. These particles emit very strong radiation at all wavelengths from short gamma rays to visible light. When used as an x-ray source, only radiation within a window of suitable wavelengths is channeled from the storage ring. **Polychromatic x-ray beams** are produced by having a broad window that allows through x-ray radiation with wavelengths of 0.2–2.0 Å. Such beams were used to record the Laue diffraction picture shown in Figure 18.2b. These very intense beams allow extremely short exposure times in diffraction experiments and can be used to collect data in experiments designed to observe changes in protein structure over very short periods of time; for example, electron transfer occurring in nanoseconds. Such studies are called **time-resolved crystallography**.

A very narrow window produces monochromatic radiation that is still several orders of magnitude more intense than the beam from conventional rotating anode x-ray sources. Such beams allow crystallographers to record diffraction patterns from very small crystals of the order of 50 micrometers or smaller. In addition, the diffraction pattern extends to higher resolution and consequently more accurate structural details are obtained as described later in this chapter. The availability and use of such beams have increased enormously in recent years and have greatly facilitated the x-ray determination of protein structures.



In diffraction experiments a narrow and parallel beam of x-rays is taken out from the x-ray source and directed onto the crystal to produce diffracted beams (Figure 18.5a). The primary beam must strike the crystal from many different directions to produce all possible diffraction spots; and so the crystal is rotated in the beam during the experiment. Rotating the crystal is much easier than rotating the x-ray source, especially when it is a synchrotron.

The incident primary beam causes damage to both protein and solvent molecules. This produces free radicals that in turn damage other molecules in the crystal. In addition, heat is generated, especially from synchrotron radiation, and eventually the primary beam burns through the crystal. To minimize this damage the crystal in the x-ray beam can be cooled to about -150°C . Such cooling does not prevent the formation of free radicals but greatly reduces the rate at which they can diffuse in the crystal, and greatly prolongs the life of the crystal in the x-ray beam. To prevent ice forming and destroying the crystal, it is essential to replace some of the water in the crystal with cryoprotectants. The crystal is suspended in the cryoprotectant, then chilled in liquid nitrogen and transferred to the x-ray beam where it is kept in a fine jet of nitrogen gas from boiling liquid nitrogen. Cryocooling has proved to be one very important technical innovation and during recent years has been adopted by most crystallographic laboratories.

X-ray data are recorded either on image plates or by electronic detectors

Today the diffracted spots are usually recorded on an image plate rather than on x-ray film, the classical method (see Figure 18.5b), or by an electronic detector. The **image plate** is in effect a reusable film. The diffraction pattern recorded on the plate is scanned and stored in a computer. The image plate is then erased and ready for reuse. Electronic **area detectors** feed the signals they detect directly in a digitized form into a computer, and can therefore be regarded as an electronic film. They significantly reduce the time required to collect and measure diffraction data. To determine the structure of a protein, as we will see, it is necessary to compare x-ray data from native crystals of the protein with those from crystals in which different atoms of the protein are complexed with heavy metals. Moreover, to elucidate a protein's function x-ray data must also be collected from complexes with different types of bound ligands. In total, therefore, several hundred thousand diffraction spots are usually collected and measured for each protein.

Figure 18.5 Schematic view of a diffraction experiment. (a) A narrow beam of x-rays (red) is taken out from the x-ray source through a collimating device. When the primary beam hits the crystal, most of it passes straight through, but some is diffracted by the crystal. These diffracted beams, which leave the crystal in many different directions, are recorded on a detector, either a piece of x-ray film or an area detector. (b) A diffraction pattern from a crystal of the enzyme RuBisCo using monochromatic radiation (compare with Figure 18.2b, the pattern using polychromatic radiation). The crystal was rotated one degree while this pattern was recorded.

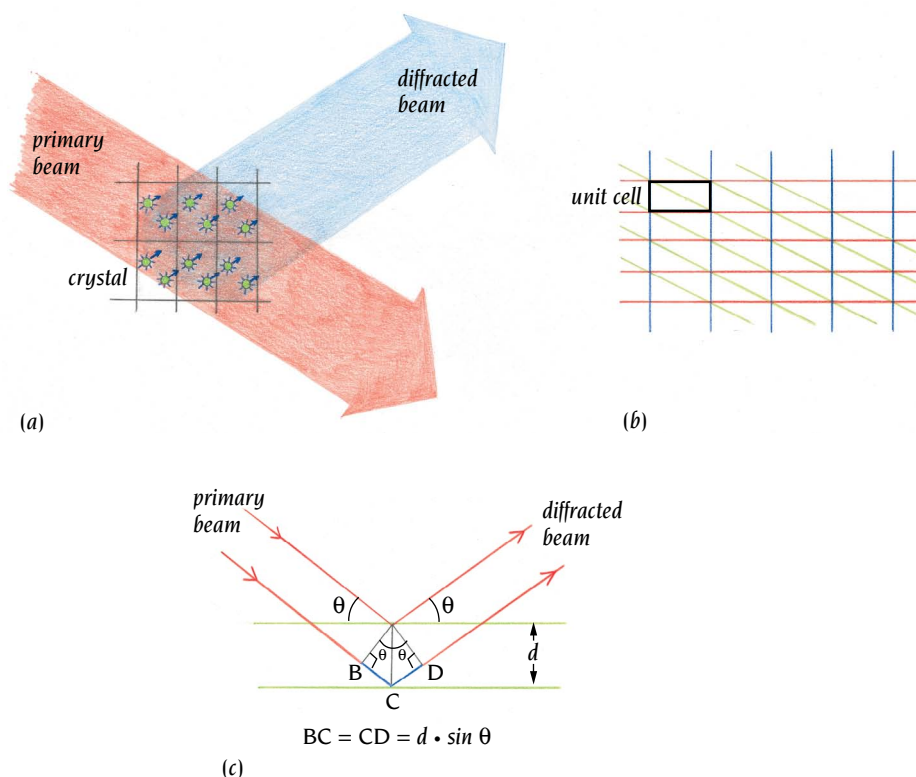


Figure 18.6 Diffraction of x-rays by a crystal. (a) When a beam of x-rays (red) shines on a crystal all atoms (green) in the crystal scatter x-rays in all directions. Most of these scattered x-rays cancel out, but in certain directions (blue arrow) they reinforce each other and add up to a diffracted beam. (b) Different sets of parallel planes can be arranged through the crystal so that each corner of all unit cells is on one of the planes of the set. The diagram shows in two dimensions three simple sets of parallel lines: red, blue, and green. A similar effect is seen when driving past a plantation of regularly spaced trees. One sees the trees arranged in different sets of parallel rows. (c) X-ray diffraction can be regarded as reflection of the primary beam from sets of parallel planes in the crystal. Two such planes are shown (green), separated by a distance d . The primary beam strikes the planes at an angle θ and the reflected beam leaves at the same angle, the reflection angle. X-rays (red) that are reflected from the lower plane have traveled farther than those from the upper plane by a distance $BC + CD$, which is equal to $2d \cdot \sin \theta$. Reflection can only occur when this distance is equal to the wavelength λ of the x-ray beam and Bragg's law— $2d \cdot \sin \theta = \lambda$ —gives the conditions for diffraction. To determine the size of the unit cell, the crystal is oriented in the beam so that reflection is obtained from the specific set of planes in which any two adjacent planes are separated by the length of one of the unit cell axes. This distance, d , is then equal to $\lambda / (2 \cdot \sin \theta)$. The wavelength, λ , of the beam is known since we use monochromatic radiation. The reflection angle, θ , can be calculated from the position of the diffracted spot on the film, using the relation derived in Figure 18.7, where the crystal to film distance can be easily measured. The crystal is then reoriented, and the procedure is repeated for the other two axes of the unit cell.

The rules for diffraction are given by Bragg's law

When the primary beam from an x-ray source strikes the crystal, most of the x-rays travel straight through it. Some, however, interact with the electrons on each atom and cause them to oscillate. The oscillating electrons serve as a new source of x-rays, which are emitted in almost all directions. We refer to this rather loosely as **scattering**. When atoms and hence their electrons are arranged in a regular three-dimensional array, as in a crystal, the x-rays emitted from the oscillating electrons interfere with one another. In most cases, these x-rays, colliding from different directions, cancel each other out; those from certain directions, however, will add together to produce diffracted beams of radiation that can be recorded as a pattern on a photographic plate or detector (Figure 18.6a).

How is the diffraction pattern obtained in an x-ray experiment such as that shown in Figure 18.5b related to the crystal that caused the diffraction? This question was addressed in the early days of x-ray crystallography by Sir Lawrence Bragg of Cambridge University, who showed that diffraction by a crystal can be regarded as the reflection of the primary beam by sets of parallel planes, rather like a set of mirrors, through the unit cells of the crystal (see Figure 18.6b and c).

X-rays that are reflected from adjacent planes travel different distances (see Figure 18.6c), and Bragg showed that diffraction only occurs when the difference in distance is equal to the wavelength of the x-ray beam. This distance is dependent on the reflection angle, which is equal to the angle between the primary beam and the planes (see Figure 18.6c).

The relationship between the reflection angle, θ , the distance between the planes, d , and the wavelength, λ , is given by **Bragg's law**: $2d \cdot \sin \theta = \lambda$. This relation can be used to determine the size of the unit cell (see legend to Figure 18.6c and Figure 18.7). Briefly, the position on the film of the diffraction data relates each spot to a specific set of planes through the crystal. By using Bragg's law, these positions can be used to determine the size of the unit cell.

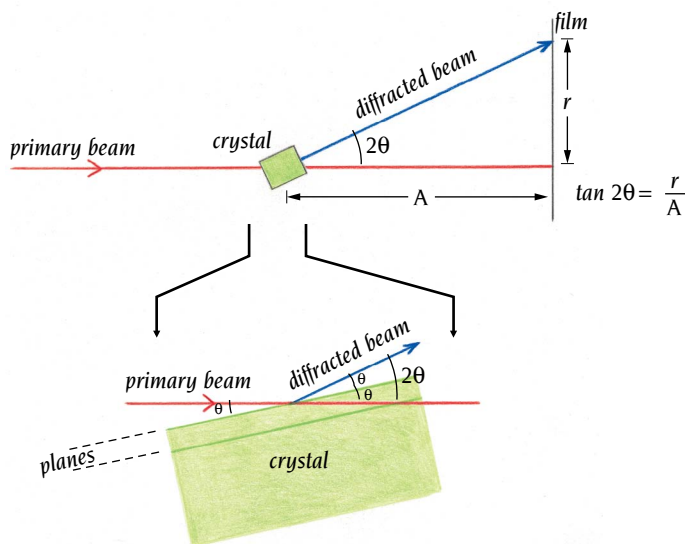


Figure 18.7 The reflection angle, θ , for a diffracted beam can be calculated from the distance (r) between the diffracted spot on a film and the position where the primary beam hits the film. From the geometry shown in the diagram the tangent of the angle $2\theta = r/A$. A is the distance between crystal and film that can be measured on the experimental equipment, while r can be measured on the film. Hence θ can be calculated. The angle between the primary beam and the diffracted beam is 2θ as can be seen on the enlarged insert at the bottom. It shows that this angle is equal to the angle between the primary beam and the reflecting plane plus the reflection angle, both of which are equal to θ .

Phase determination is the major crystallographic problem

Each atom in a crystal scatters x-rays in all directions, and only those that positively interfere with one another, according to Bragg's law, give rise to diffracted beams (see Figure 18.6a) that can be recorded as a distinct **diffraction spot** above background. Each diffraction spot is the result of interference of all x-rays with the same diffraction angle emerging from all atoms. For a typical protein crystal, myoglobin, each of the about 20,000 diffracted beams that have been measured contains scattered x-rays from each of the around 1500 atoms in the molecule. To extract information about individual atoms from such a system requires considerable computation. The mathematical tool that is used to handle such problems is called the **Fourier transform**, invented by the French mathematician Jean Baptiste Joseph Fourier while he served as a bureaucrat in the government of Napoleon Bonaparte.

Each diffracted beam, which is recorded as a spot on the film, is defined by three properties: the **amplitude**, which we can measure from the intensity of the spot; the **wavelength**, which is set by the x-ray source; and the **phase**, which is lost in x-ray experiments (Figure 18.8). We need to know all three properties for all of the diffracted beams to determine the position of the atoms giving rise to the diffracted beams. How do we find the phases of the diffracted beams? This is the so-called phase problem in x-ray crystallography.

In small-molecule crystallography the phase problem was solved by so-called direct methods (recognized by the award of a Nobel Prize in chemistry to Jerome Karle, US Naval Research Laboratory, Washington, DC, and Herbert Hauptman, the Medical Foundation, Buffalo). For larger molecules, protein crystallographers have stayed at the laboratory bench using a method pioneered by Max Perutz and John Kendrew and their co-workers to circumvent the phase problem. This method, called **multiple isomorphous replacement**

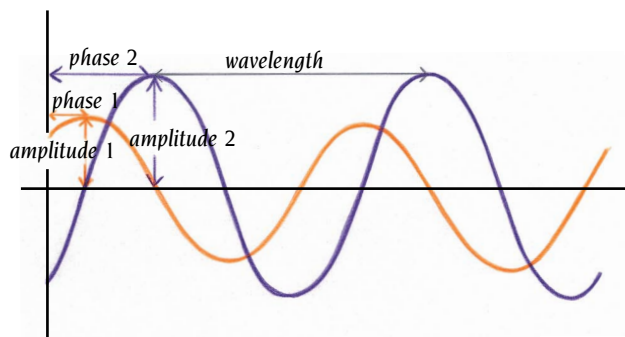


Figure 18.8 Two diffracted beams (purple and orange), each of which is defined by three properties: amplitude, which is a measure of the strength of the beam and which is proportional to the intensity of the recorded spot; phase, which is related to its interference, positive or negative, with other beams; and wavelength, which is set by the x-ray source for monochromatic radiation.

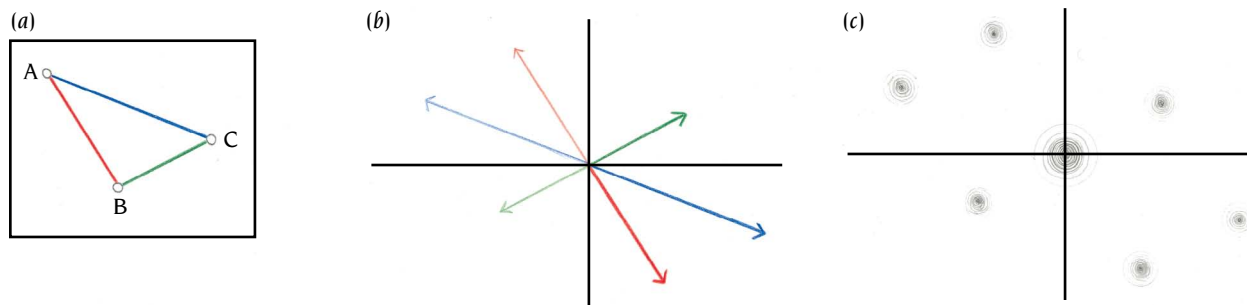


Figure 18.9 Fourier summations of the intensity differences between diffracted spots from crystals of the protein alone and protein plus heavy metals give vector maps between the heavy atoms. Three atoms—A, B, and C—are at specific positions in the unit cell in (a). They give vectors A–B, A–C, and B–C, which are drawn from a common origin in (b) in dark colors. They also give the same vectors in the opposite directions as shown in light colors. The experimentally observed vector map is shown in (c) with a large peak at the origin corresponding to zero vectors between an atom and itself. It is straightforward to deduce the map in (c) from the atomic arrangement in (a). It is more difficult to do the reverse, to deduce the atomic arrangement in (a) from the vector map in (c), especially if there are many atoms in the unit cell that give rise to a large number of peaks in the vector map. For example, with 10 atoms in the unit cell there are 90 different vectors between the atoms.

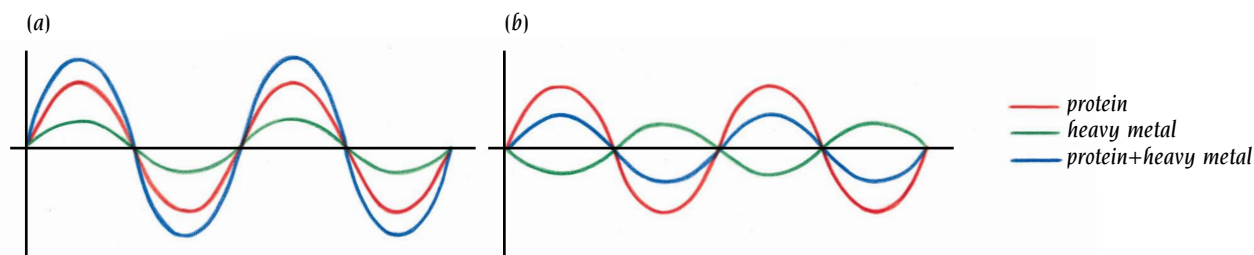
(MIR), requires the introduction of new x-ray scatterers into the unit cell of the crystal. These additions should be heavy atoms (so that they make a significant contribution to the diffraction pattern); there should not be too many of them (so that their positions can be located); and they should not change the structure of the molecule or of the crystal cell—in other words, the crystals should be isomorphous. In practice, isomorphous replacement is usually done by diffusing different heavy-metal complexes into the channels of preformed protein crystals. With luck the protein molecules expose side chains in these solvent channels, such as SH groups, that are able to bind heavy metals. It is also possible to replace endogenous light metals in metalloproteins with heavier ones, e.g., zinc by mercury or calcium by samarium.

Since such heavy metals contain many more electrons than the light atoms, H, N, C, O, and S, of the protein, they scatter x-rays more strongly. All diffracted beams would therefore increase in intensity after heavy-metal substitution if all interference were positive. In fact, however, some interference is negative; consequently, following heavy-metal substitution, some spots measurably increase in intensity, others decrease, and many show no detectable difference.

How do we find phase differences between diffracted spots from intensity changes following heavy-metal substitution? We first use the intensity differences to deduce the positions of the heavy atoms in the crystal unit cell. Fourier summations of these intensity differences give maps of the vectors between the heavy atoms, the so-called **Patterson maps** (Figure 18.9). From these vector maps it is relatively easy to deduce the atomic arrangement of the heavy atoms, so long as there are not too many of them. From the positions of the heavy metals in the unit cell, one can calculate the amplitudes and phases of their contribution to the diffracted beams of the protein crystals containing heavy metals.

How is that knowledge used to find the phase of the contribution from the protein in the absence of the heavy-metal atoms? We know the phase and amplitude of the heavy metals and the amplitude of the protein alone. In addition, we know the amplitude of protein plus heavy metals (i.e., protein heavy-metal complex); thus we know one phase and three amplitudes. From this we can calculate whether the interference of the x-rays scattered by the heavy metals and protein is constructive or destructive (Figure 18.10). The extent of positive or negative interference plus knowledge of the phase of the heavy metal together give an estimate of the phase of the protein.

Figure 18.10 The diffracted waves from the protein part (red) and from the heavy metals (green) interfere with each other in crystals of a heavy-atom derivative. If this interference is positive as illustrated in (a), the intensity of the spot from the heavy-atom derivative (blue) crystal will be stronger than that of the protein (red) alone (larger amplitude). If the interference is negative as in (b), the reverse is true (smaller amplitude).



Unfortunately, the problem is underdetermined so that two different phase angles are equally good solutions. To distinguish between these two possible solutions, a second heavy-metal complex must be used, which also gives two possible phase angles. Only one of these will have the same value as one of the two previous phase angles; it therefore represents the correct phase angle. In practice, more than two different heavy-metal complexes are needed to give a reasonably good phase determination for all reflections. Each individual phase estimate contains experimental errors arising from errors in the measured amplitudes; furthermore, for many reflections, the intensity differences are too small to measure after one particular isomorphous replacement, and others must be tried.

Phase information can also be obtained by Multiwavelength Anomalous Diffraction experiments

For certain x-ray wavelengths, the interaction between the x-rays and the electrons of an atom causes the electrons to absorb the energy of the x-ray. This causes a change in the x-ray scattering of the atom, called **anomalous scattering**, that depends both on the type of atom and on the wavelength of the x-rays. The size of this change is small and negligible for light atoms such as hydrogen, carbon, nitrogen and oxygen but is measurable for heavier atoms such as selenium, iron, zinc, and mercury. Due to this effect, small changes of the wavelength of the incident x-ray beam around the absorption edge of the heavy atom produce measurable intensity differences in the diffraction pattern. It is sufficient to have only one such heavy atom bound to each protein molecule of medium size (about 200 amino acid residues) for the effect to be measurable.

The intensity differences obtained in the diffraction pattern by illuminating such a crystal by x-rays of different wavelengths can be used in a way similar to the method of multiple isomorphous replacement to obtain the phases of the diffracted beams. This method of phase determination which is called **Multiwavelength Anomalous Diffraction, MAD**, and which was pioneered by Wayne Hendrickson at Columbia University, US, is now increasingly used by protein crystallographers.

The MAD method requires access to synchrotron radiation since different wavelengths are used, and it also requires that the crystal contains heavy atoms. Some protein molecules, such as metalloenzymes, contain intrinsic metal atoms but most proteins do not. However, using recombinant DNA technology it is possible to incorporate selenomethionine instead of methionine into recombinant proteins, thereby fulfilling the requirements for using the MAD method. Proteins with selenomethionine have very similar structures to the methionine-containing proteins. The structure of a number of selenomethionine-containing proteins, including several described in earlier chapters, has been determined using data collected at experimental stations specifically designed for MAD experiments at several synchrotron sources. Alternatively, proteins can be soaked in heavy metal solutions as for conventional x-ray structure determination (as discussed above).

Building a model involves subjective interpretation of the data

The amplitudes and the phases of the diffraction data from the protein crystals are used to calculate an **electron-density map** of the repeating unit of the crystal. This map then has to be interpreted as a polypeptide chain with a particular amino acid sequence. The interpretation of the electron-density map is complicated by several limitations of the data. First of all, the map itself contains errors, mainly due to errors in the phase angles. In addition, the quality of the map depends on the **resolution** of the diffraction data, which in turn depends on how well-ordered the crystals are. This directly influences the image that can be produced. The resolution is measured in Å

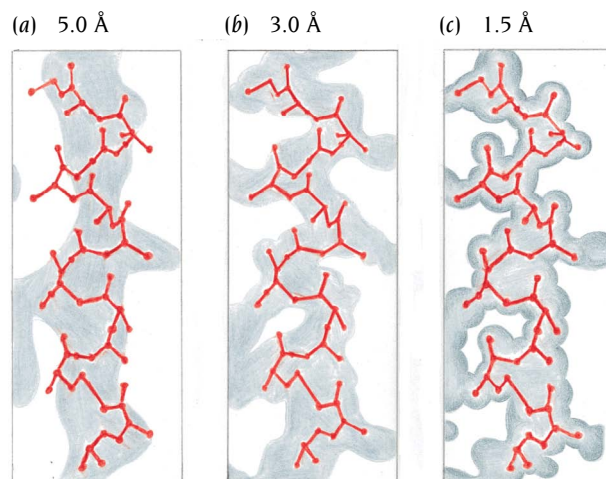


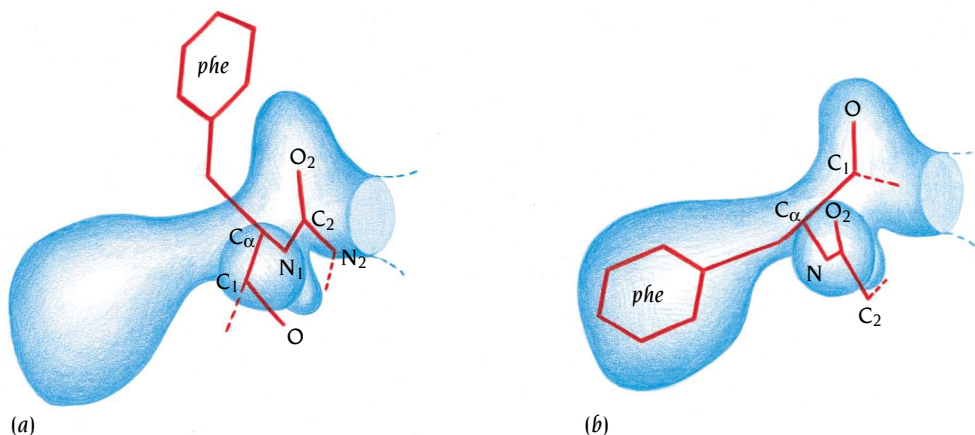
Figure 18.11 Electron-density maps at different resolution show more detail at higher resolution. (a) At low resolution (5.0 Å) individual groups of atoms are not resolved, and only the rodlike feature of an α helix can be deduced. (b) At medium resolution (3.0 Å) the path of the polypeptide chain can be traced, and (c) at high resolution (1.5 Å) individual atoms start to become resolved. Relevant parts of the protein chain (red) are superimposed on the electron densities (gray). The diagrams show one α helix from a small protein, myohemerythrin. [Adapted from W.A. Hendrickson in *Protein Engineering* (eds. D.L. Oxender and C.F. Fox.), p. 11. New York: Liss, 1987.]

units; the smaller this number is, the higher the resolution and therefore the greater the amount of detail that can be seen (Figure 18.11).

From a map at low resolution (5 Å or higher) one can obtain the shape of the molecule and sometimes identify α -helical regions as rods of electron density. At medium resolution (around 3 Å) it is usually possible to trace the path of the polypeptide chain and to fit a known amino acid sequence into the map. At this resolution it should be possible to distinguish the density of an alanine side chain from that of a leucine, whereas at 4 Å resolution there is little side chain detail. Gross features of functionally important aspects of a structure usually can be deduced at 3 Å resolution, including the identification of active-site residues. At 2 Å resolution details are sufficiently well resolved in the map to decide between a leucine and an isoleucine side chain, and at 1 Å resolution one sees atoms as discrete balls of density. However, the structures of only a few small proteins have been determined to such high resolution.

Building the initial **model** is a trial-and-error process. First, one has to decide how the polypeptide chain weaves its way through the electron-density map. The resulting chain trace constitutes a hypothesis, by which one tries to match the density of the side chains to the known sequence of the polypeptide. This sounds easy, but it is not; a map showing continuous density from N-terminus to C-terminus is rare. More usually one produces a number of matches between the electron density and discontinuous regions of the sequence that may initially account for only a small fraction of the molecule and may be internally inconsistent. When a reasonable chain trace has finally been obtained, an initial model is built to give the best fit of the atoms to the electron density (Figure 18.12). Today, computer graphics are exploited both for chain tracing and for model building to present the data and manipulate the models.

Figure 18.12 The electron-density map is interpreted by fitting into it pieces of a polypeptide chain with known stereochemistry such as peptide groups and phenyl rings. The electron density (blue) is displayed on a graphics screen in combination with a part of the polypeptide chain (red) in an arbitrary orientation (a). The units of the polypeptide chain can then be rotated and translated relative to the electron density until a good fit is obtained (b). Notice that individual atoms are not resolved in such electron densities, there are instead lumps of density corresponding to groups of atoms. [Adapted from A. Jones *Methods Enzym.* (eds. H.W. Wyckoff, C.H. Hirs, and S.N. Timasheff) 115B: 162, New York: Academic Press, 1985.]



Errors in the initial model are removed by refinement

The initial model will contain errors. Provided the protein crystals diffract to high enough resolution (better than ~ 2.5 Å), most of the errors can be removed by crystallographic refinement of the model. In this process the model is changed to minimize the difference between the experimentally observed diffraction amplitudes and those calculated for a hypothetical crystal containing the model instead of the real molecule. This difference is expressed as an **R factor**, residual disagreement, which is 0.0 for exact agreement and around 0.59 for total disagreement.

In general, the R factor is between 0.15 and 0.20 for a well-determined protein structure. The residual difference rarely is due to large errors in the model of the protein molecule, but rather it is an inevitable consequence of errors and imperfections in the data. These derive from various sources, including slight variations in conformation of the protein molecules and inaccurate corrections both for the presence of solvent and for differences in the orientation of the microcrystals from which the crystal is built. This means that the final model represents an average of molecules that are slightly different both in conformation and orientation, and not surprisingly the model never corresponds precisely to the actual crystal.

The atoms of a protein's structure are usually defined by four parameters, three coordinates that give their position in space and one quantity, B, which is called the temperature factor. For well refined, correct structures these B-values are of the order of 20 or less. High B-values, 40 or above, in a local region can be due to flexibility or slight disorder, but also serve as a warning that the model of this region may be incorrect.

In refined structures at high resolution (around 2 Å) there are usually no major errors in the orientation of individual residues, and the estimated errors in atomic positions are around 0.1–0.2 Å provided the amino acid sequence is known. Hydrogen bonds both within the protein and to bound ligands can be identified with a high degree of confidence.

At medium resolution (around 3 Å) it is possible to make serious errors in the interpretation of the electron-density map, and there are, unfortunately, a number of them in the literature. Errors usually arise because elements of secondary structure are wrongly connected by the loop regions. Alpha helices and β strands in the interior of the protein are rigid in structure and well defined in the electron-density map. The loop regions, however, are usually more flexible, and therefore the corresponding electron density is less well defined. It is easy to make errors in such regions in the preliminary interpretations of electron-density maps at medium resolution. These errors are usually caught and corrected before publication since such models will not refine properly and are likely to be incompatible with existing biochemical data.

However, some models containing serious errors have been published. They all have been based on data to only medium resolution together with insufficient phase information, which gives large errors in the electron density. It should, therefore, be kept in mind that unrefined structures with R values higher than 0.30 at medium resolution may contain errors, although the overwhelming majority of such published structures have survived subsequent refinement at high resolution.

Recent technological advances have greatly influenced protein crystallography

In the early days of protein crystallography the determination of a protein structure was laborious and time consuming. The diffracted beams were obtained from weak x-ray sources and recorded on films that had to be manually scanned and measured. The available computers were far from adequate for the problem, with a computing power roughly equal to present-day pocket calculators. Computer graphics were not available, and models of the protein had to be built manually from pieces of steel rod. To determine the

structure of even a small protein molecule, therefore, required many years of work and entailed time-consuming bottlenecks at almost every stage.

The situation is radically different today. The diffraction pattern can now be recorded on electronic area detectors coupled to powerful microcomputers that immediately interpret and process the recorded signals. Data collection that only a few years ago required many months of work is now done in a few days. If the in-house x-ray source is too weak for the problem, there are synchrotron sources available in several centers around the world that provide x-ray beams that are brighter by several orders of magnitude. Powerful computers in the laboratory provide the crystallographer with immediate access to almost all the computing power he or she needs. The electron-density maps are interpreted, and models of the protein molecules are built by the crystallographer sitting in front of a computer graphics screen. He or she is greatly aided by sophisticated software that involves semiautomatic methods for the model building using knowledge from databases of previously determined and refined protein structures.

These technical advances have greatly facilitated the use of crystallography for protein structure determination. One significant problem, however, still remains: obtaining crystals that diffract to high resolution. Some protein molecules give excellent crystals after the first few trials, others may require several months of screening for the proper crystallization conditions, and many have so far resisted all attempts to crystallize them. Fortunately, it is now possible to determine the structure of small protein molecules in solution by nuclear magnetic resonance (NMR) methods, and of large complexes by a combination of x-ray or NMR studies of individual or smaller pieces and fiber diffraction or electron microscopy studies of the complete complex.

X-ray diffraction can be used to study the structure of fibers as well as crystals

As described in Chapter 2, the first complete protein structure to be determined was the globular protein myoglobin. However, the α helix that was recognized in this structure, and which has emerged as a persistent structural motif in the many hundreds of globular proteins determined subsequently, was first observed in x-ray diffraction studies of fibrous proteins.

Polymer molecules that have a high degree of regularity in their monomer sequence tend to assume helical rather than globular conformations, and fibers are formed when these helices become aligned with each other. This is well illustrated by the proteins keratin and collagen and by the nucleic acid, DNA. The regularity in the DNA double-helix is so high that fibers drawn from a concentrated DNA gel are highly ordered; the long, thread-like DNA molecules extend parallel to the length of the fiber with a high degree of regularity in their side-by-side packing extending over many molecules. These regularly packed molecules form structures known as **crystallites**, and a typical fiber of 100 microns diameter contains a large number of such crystallites separated by less ordered regions where the molecules, while still largely parallel to the fiber length, are much less regularly packed. Because the crystallites are in random orientation about the direction of the fiber length, a diffraction pattern recorded from the fiber is similar to the pattern obtained when a single crystal is rotated 360° about the vertical axis while the data is being recorded. Therefore the diffraction pattern from a crystalline fiber can be analyzed using standard crystallographic techniques. The power of crystalline fiber diffraction analysis is illustrated by the detailed stereochemical information on the A and B conformations of DNA, reviewed in Chapter 7.

The polymer molecules in fibers typically assume helical structures with one pitch of the helix forming the repeating unit. This symmetry is the origin of the characteristic “cross-like” variation in overall intensity across the diffraction pattern from helical molecules. As the degree of regularity in the arrangement of repeating units in an array decreases, the diffraction spots become broader. For a completely irregular array, such as a crystalline powder,

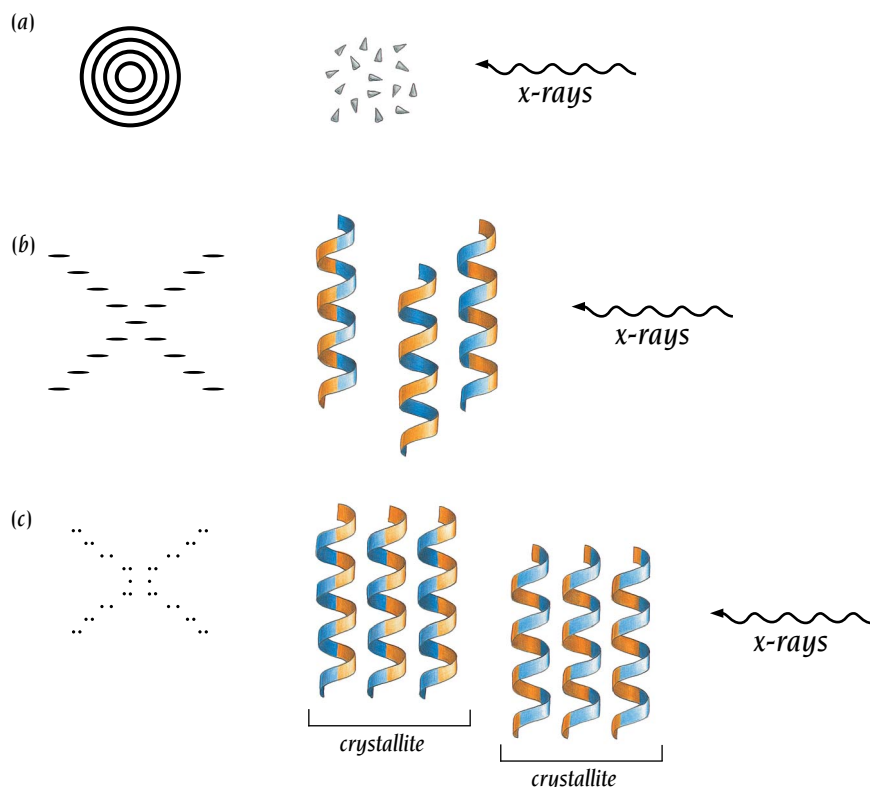
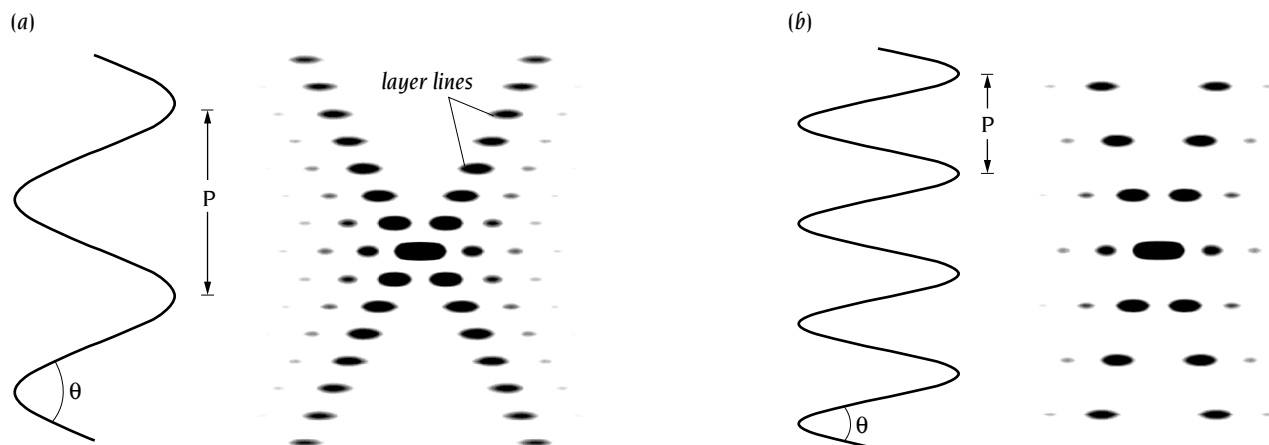


Figure 18.13 Typical fiber diffraction patterns (left) for differently ordered arrays of molecules (right). (a) A randomly oriented array, such as a powder of small crystals. The molecules within each crystal are highly ordered, but the crystals are in random orientation with respect to each other. (b) A fiber formed from an array of poorly ordered molecules. Three molecules within the array are shown (in a typical x-ray specimen there would be many more); they are parallel but have little regularity in their side-by-side packing. (c) A fiber consisting of crystallites of well ordered molecules. Two crystallites are shown, each with only three of the many molecules they would typically contain. The molecules within each crystallite are aligned with a high degree of order in their side-by-side packing and relative orientation, but the crystallites are rotated with respect to each other. A similar pattern is obtained for a single crystal that is rotated during the data collection.

the diffraction pattern reduces to the scattering from a single molecule averaged over all orientations and is observed as a continuous distribution without any characteristic diffraction spots (Figure 18.13a). However, the chains of extended helical molecules in a fiber tend to remain parallel even when there is no regularity in the orientation about the helix axis. This parallelism is reflected in the overall intensity distribution within the diffraction pattern and in particular the restriction of diffraction to layer lines (see Figure 18.13b,c). The main features in the cross-like diffraction pattern of a helical structure can be illustrated by considering diffraction by a single slit that has the shape of a sine curve and that therefore corresponds to a projection of a helix. The diffraction patterns for projections of two helices differing in pitch length are illustrated in Figure 18.14. The separation between the layer lines is determined by the helix pitch: as the helix pitch increases the layer lines move closer together (compare parts a and b of Figure 18.14).

In addition to the reciprocal relationship between the helix pitch and layer line spacing, Figure 18.14 illustrates the reciprocal relationship between the orientation of the arms of the cross and the angle of climb of the helix: as the helix becomes steeper the arms of the cross become more horizontal.

Figure 18.14 The diffraction pattern of helices in fiber crystallites can be simulated by the diffraction pattern of a single slit with the shape of a sine curve (representing the projection of a helix). Two such simulations are given in (a) and (b), with the helix shown to the left of its diffraction pattern. The spacing between the layer lines is inversely related to the helix pitch, P and the angle of the cross arms in the diffraction pattern is related to the angle of climb of the helix, θ . The helix in (b) has a smaller pitch and angle of climb than the helix in (a). (Courtesy of W. Fuller.)



The structure of biopolymers can be studied using fiber diffraction

A number of important protein molecules occur *in vivo* as fibrous structures. In keratin and collagen, groups of extended polymer molecules are wound around each other like strands in a rope, while muscle fibers contain a mixture of extended and globular components in a highly ordered array (see Chapter 14). The periodicities in these complex arrays of polymer molecules may be as large as hundreds or thousands of angstroms, whereas the periodicities of structural features in individual polymer molecules such as the pitch of the DNA double helix are typically a few tens of angstroms. The fibrous state offers a less constrained environment than a single crystal so that it is possible to follow changes in the x-ray fiber diffraction pattern as a consequence of structural transitions. In the case of DNA, changes of helix pitch and the number of nucleotide-pairs per pitch have been followed, while in muscle such time-resolved techniques have allowed changes in the diffraction pattern to be recorded during the contractile cycle.

The diffraction from helical molecules can be illustrated by the diffraction patterns from DNA (Figure 18.15). The pattern from the A form on the left has sharp diffraction spots across the whole pattern, indicating the regular packing of molecules in crystallites. In contrast, the pattern from the B form consists almost entirely of continuous diffraction along layer lines and the structure is said to be semi-crystalline. The pitch of the A form is 28 Å compared with 34 Å for the B form and this is reflected in Figure 18.15 by the smaller spacing between layer lines in the B form. The strong meridional diffraction near the top of the pattern from the B form is due to scattering from the stack of base pairs that form the core of the DNA double helix. Its position indicates that the distance between successive base pairs is 3.4 Å and therefore that there are 10 base pairs per helix pitch.

Knowing the helix pitch, it is possible to determine an approximate value for the radius of the helix in the B form from the inclination of the

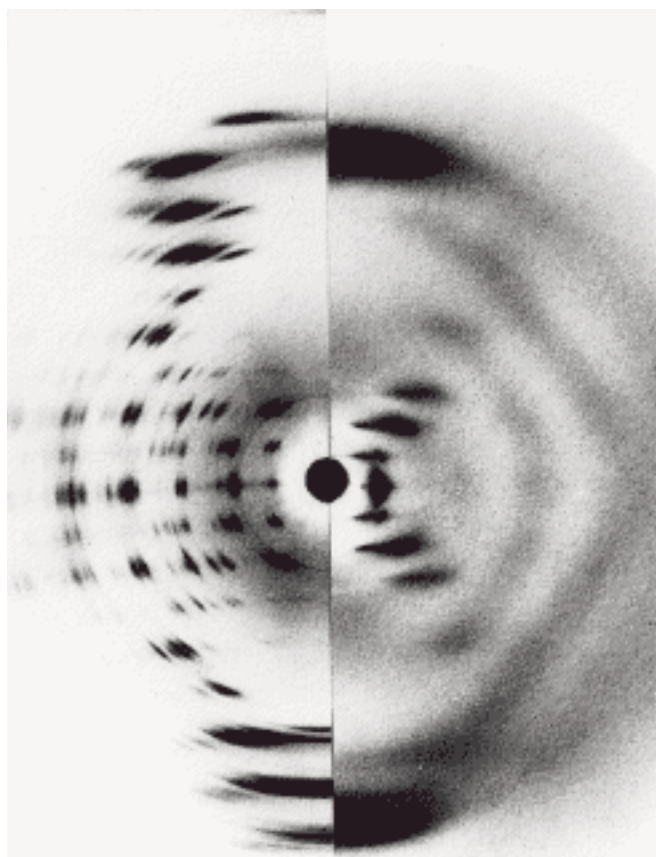
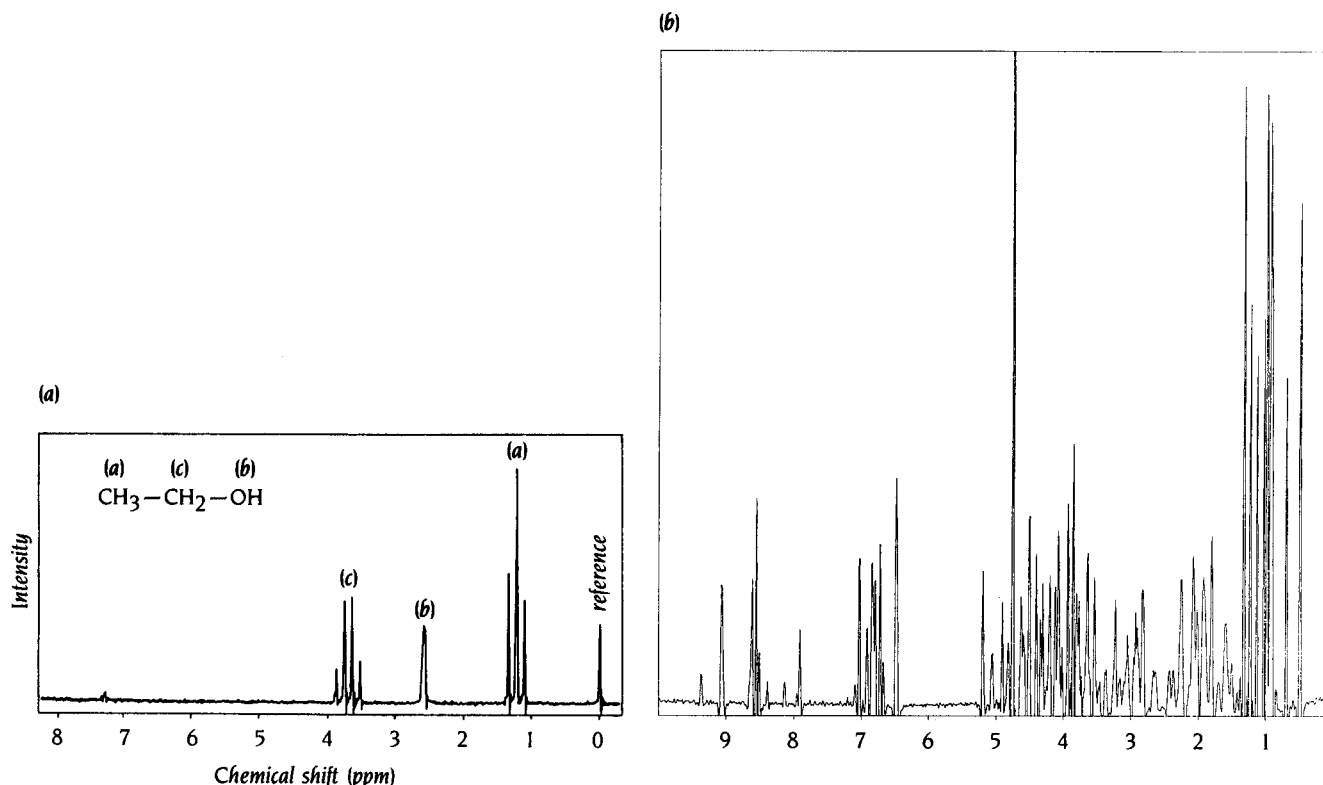


Figure 18.15 Diffraction patterns of DNA, showing the patterns obtained for both A-DNA (left half) and B-DNA (right half). (Courtesy of W. Fuller.)



arms of the cross in the B diffraction pattern. Arguments such as those outlined above, based on x-ray fiber diffraction patterns obtained by Maurice Wilkins and Rosalind Franklin, were used by Jim Watson and Francis Crick in the construction of their double-helical model for DNA.

NMR methods use the magnetic properties of atomic nuclei

Certain atomic nuclei, such as ^1H , ^{13}C , ^{15}N , and ^{31}P have a magnetic moment or **spin**. The chemical environment of such nuclei can be probed by **nuclear magnetic resonance (NMR)** and this technique can be exploited to give information on the distances between atoms in a molecule. These distances can then be used to derive a three-dimensional model of the molecule. Most structure determinations of protein molecules by NMR have used the spin of ^1H , since hydrogen atoms are abundant in proteins. Small proteins can be analyzed by ^1H (proton) NMR but to study larger proteins and to obtain sufficient data to determine side chain conformations it is necessary to introduce ^{13}C and ^{15}N into the protein. This is usually done by producing the protein in microorganisms grown in media enriched with these isotopes. NMR studies of proteins containing one of the isotopes are called 3-D NMR, and when both ^{13}C and ^{15}N are present they are called 4-D NMR.

When protein molecules are placed in a strong magnetic field, the spin of their hydrogen atoms aligns along the field. This equilibrium alignment can be changed to an excited state by applying **radio frequency (RF)** pulses to the sample. When the nuclei of the protein molecule revert to their equilibrium state, they emit RF radiation that can be measured. The exact frequency of the emitted radiation from each nucleus depends on the molecular environment of the nucleus and is different for each atom, unless they are chemically equivalent and have the same molecular environment (Figure 18.16a). These different frequencies are obtained relative to a reference signal and are called **chemical shifts**. The nature, duration, and combination of applied RF pulses can be varied enormously, and different molecular properties of the sample can be probed by selecting the appropriate combination of pulses.

Figure 18.16 One-dimensional NMR spectra.

(a) ^1H -NMR spectrum of ethanol. The NMR signals (chemical shifts) for all the hydrogen atoms in this small molecule are clearly separated from each other. In this spectrum the signal from the CH_3 protons is split into three peaks and that from the CH_2 protons into four peaks close to each other, due to the experimental conditions. (b) ^1H -NMR spectrum of a small protein, the C-terminal domain of a cellulase, comprising 36 amino acid residues. The NMR signals from many individual hydrogen atoms overlap and peaks are obtained that comprise signals from many hydrogen atoms. (Courtesy of Per Kraulis, Uppsala, from data published in Kraulis et al., *Biochemistry* 28: 7241–7257, 1989.)

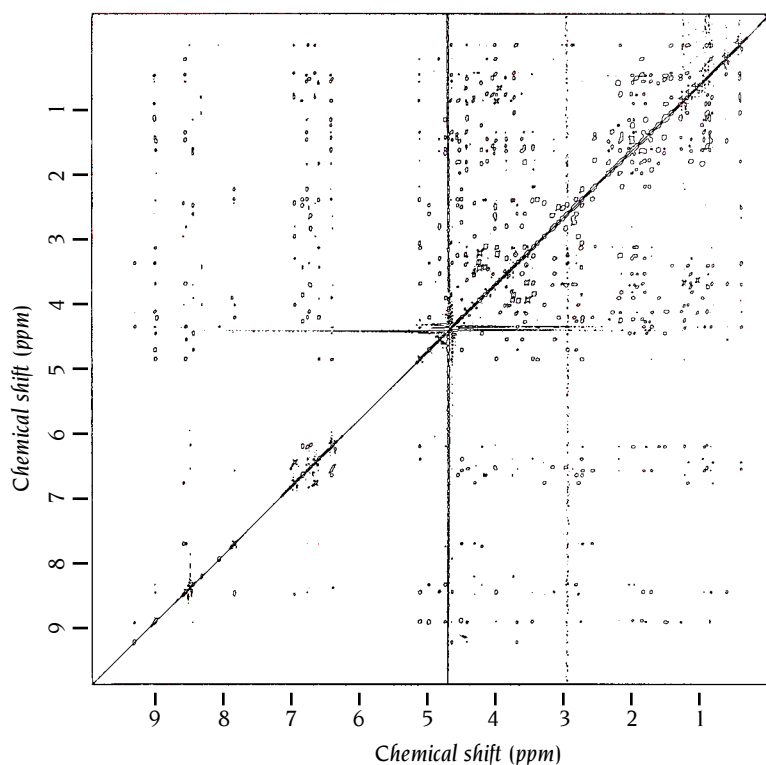


Figure 18.17 Two-dimensional NMR spectrum of the C-terminal domain of a cellulase. The peaks along the diagonal correspond to the spectrum shown in Figure 18.16b. The off-diagonal peaks in this NOE spectrum represent interactions between hydrogen atoms that are closer than 5 Å to each other in space. From such a spectrum one can obtain information on both the secondary and tertiary structures of the protein. (Courtesy of Per Kraulis, Uppsala.)

In principle, it is possible to obtain a unique signal (chemical shift) for each hydrogen atom in a protein molecule, except those that are chemically equivalent, for example, the protons on the CH_3 side chain of an alanine residue. In practice, however, such one-dimensional NMR spectra of protein molecules (see Figure 18.16b) contain overlapping signals from many hydrogen atoms because the differences in chemical shifts are often smaller than the resolving power of the experiment. In recent years this problem has been bypassed by designing experimental conditions that yield a two-dimensional NMR spectrum, the results of which are usually plotted in a diagram as shown in Figure 18.17.

The diagonal in such a diagram corresponds to a normal one-dimensional NMR spectrum. The peaks off the diagonal result from interactions between hydrogen atoms that are close to each other in space. By varying the nature of the applied RF pulses these off-diagonal peaks can reveal different types of interactions. A COSY (correlation spectroscopy) experiment gives peaks between hydrogen atoms that are covalently connected through one or two other atoms, for example, the hydrogen atoms attached to the nitrogen and C_α atoms within the same amino acid residue (Figure 18.18a). An NOE (nuclear Overhauser effect) spectrum, on the other hand, gives peaks between pairs of hydrogen atoms that are close together in space even if they are from amino acid residues that are quite distant in the primary sequence (see Figure 18.18b).

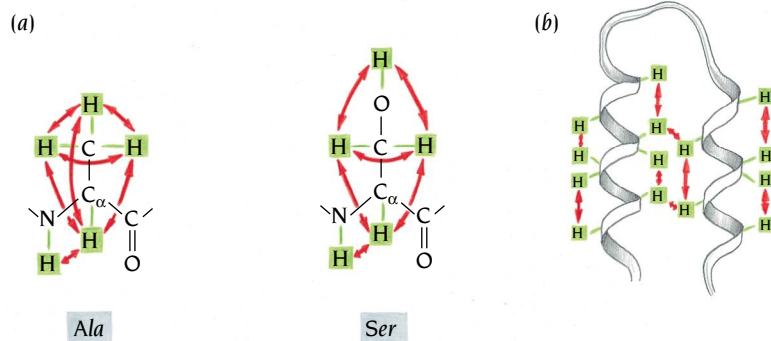


Figure 18.18 (a) COSY NMR experiments give signals that correspond to hydrogen atoms that are covalently connected through one or two other atoms. Since hydrogen atoms in two adjacent residues are covalently connected through at least three other atoms (for instance, $\text{HC}_\alpha\text{-C}'\text{-NH}$), all COSY signals reveal interactions within the same amino acid residue. These interactions are different for different types of side chains. The NMR signals therefore give a “fingerprint” of each amino acid. The diagram illustrates fingerprints (red) of residues Ala and Ser. (b) NOE NMR experiments give signals that correspond to hydrogen atoms that are close together in space (less than 5 Å), even though they may be far apart in the amino acid sequence. Both secondary and tertiary structures of small protein molecules can be derived from a collection of such signals, which define distance constraints between a number of hydrogen atoms along the polypeptide chain.

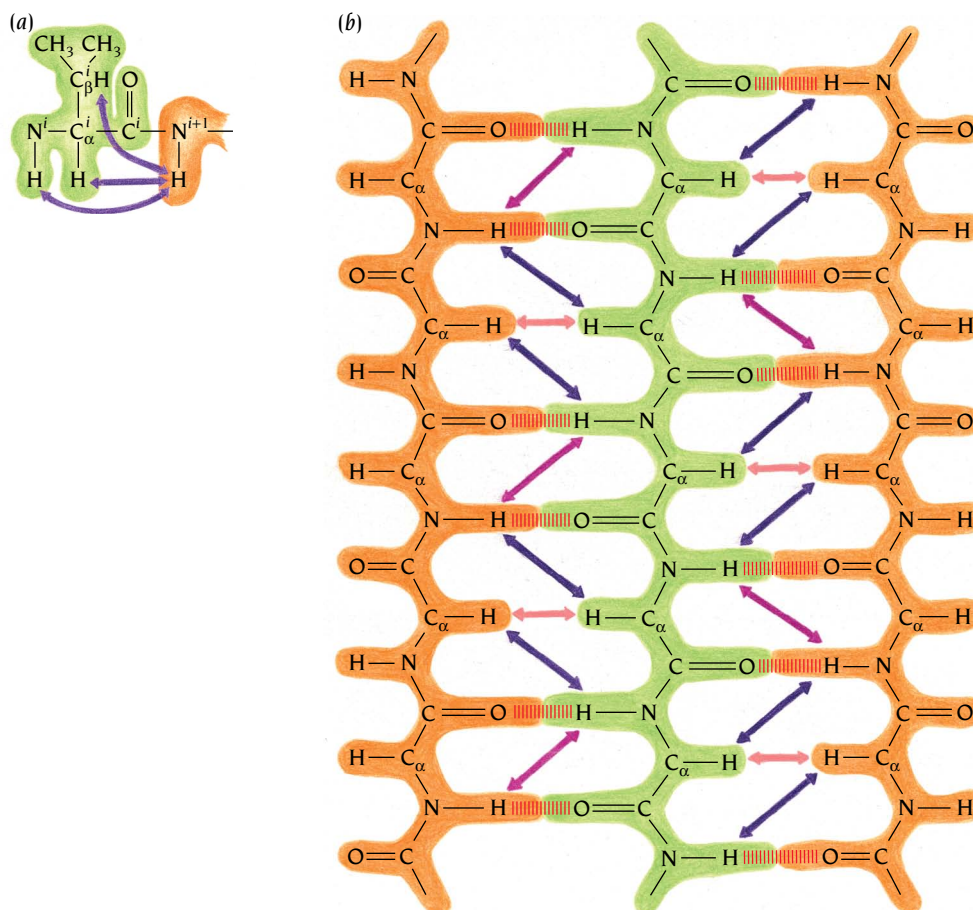
Two-dimensional NMR spectra of proteins are interpreted by the method of sequential assignment

Two-dimensional NOE spectra, by specifying which groups are close together in space, contain three-dimensional information about the protein molecule. It is far from trivial, however, to assign the observed peaks in the spectra to hydrogen atoms in specific residues along the polypeptide chain because the order of peaks along the diagonal has no simple relation to the order of amino acids along the polypeptide chain. This problem has in principle been solved in the laboratory of Kurt Wüthrich in the ETH, Zürich, where the method of **sequential assignment** was developed.

Sequential assignment is based on the differences in the number of hydrogen atoms and their covalent connectivity in the different amino acid residues. Each type of amino acid has a specific set of covalently connected hydrogen atoms that will give a specific combination of **cross-peaks**, a “fingerprint,” in a COSY spectrum (see Figure 18.18a). From the COSY spectrum it is therefore possible to identify the H atoms that belong to each amino acid residue and, in addition, determine the nature of the side chain of that residue. However, the order of these fingerprints along the diagonal has no relation to the amino acid sequence of the protein. For example, when the fingerprint in one specific region of the COSY spectrum of the *lac*-repressor segment was assigned to a Ser residue, it was not known whether this fingerprint corresponded to Ser 16, Ser 28, or Ser 31 in the amino acid sequence.

The sequence-specific assignment, however, can be made from NOE spectra (see Figures 18.17 and 18.18b) that record signals from H atoms that are close together in space. In addition to the interactions between H atoms that are far apart in the sequence, these spectra also record interactions between H atoms from sequentially adjacent residues, specifically, interactions from the H atom attached to the main chain N of residue number $i + 1$ to H atoms bonded to N, C_{α} , and C_{β} of residue number i (Figure 18.19a).

Figure 18.19 (a) Adjacent residues in the amino acid sequence of a protein can be identified from NOE spectra. The H atom attached to residue $i + 1$ (orange) is close to and interacts with (purple arrows) the H atoms attached to N, C_{α} and C_{β} of residue i (light green). These interactions give cross-peaks in the NOE spectrum that identify adjacent residues and are used for sequence-specific assignment of the amino acid fingerprints derived from a COSY spectrum. (b) Regions of secondary structure in a protein have specific interactions between hydrogen atoms in sequentially nonadjacent residues that give a characteristic pattern of cross-peaks in an NOE spectrum. In antiparallel β -sheet regions there are interactions between C_{α} -H atoms of adjacent strands (pink arrows), between N-H and C_{α} -H atoms (dark purple arrows), and between N-H atoms of adjacent strands (light purple arrows). The corresponding pattern of cross-peaks in an NOE spectrum identifies the residues that form the antiparallel β sheet. Parallel β sheets and α helices are identified in a similar way.



These signals in the NOE spectra therefore in principle make it possible to determine which fingerprint in the COSY spectrum comes from a residue adjacent to the one previously identified. For example, in the case of the *lac*-repressor fragment the specific Ser residue that was identified from the COSY spectrum was shown in the NOE spectrum to interact with a His residue, which in turn interacted with a Val residue. Comparison with the known amino acid sequence revealed that the tripeptide Ser-His-Val occurred only once, for residues 28–30.

In practice, it is difficult to make unique assignments for longer pieces than di- or tri-peptides, since NOE signals also occur between residues close together in space but far apart in the sequence. Therefore, the peptide segments that have been uniquely identified by NMR are usually matched with corresponding segments in the independently determined amino acid sequence of the protein. Thus knowledge of the amino acid sequence is just as essential for the correct interpretation of NMR spectra as it is for the interpretation of electron-density maps in x-ray crystallography. Whereas x-ray crystallography directly gives an image of a three-dimensional model of the protein molecule, NMR spectroscopy identifies H atoms in the protein that are close together in space, and this information is then used to derive, indirectly, a three-dimensional model of the protein.

Distance constraints are used to derive possible structures of a protein molecule

The final result of the sequence-specific assignment of NMR signals, preferably done using interactive computer graphics, is a list of **distance constraints** from specific hydrogen atoms in one residue to hydrogen atoms in a second residue. The list contains a large number of such distances, which are usually divided into three intervals within the region 1.8 Å to 5 Å, depending on the intensity of the NOE peak. This list immediately identifies the secondary structure elements of the protein molecule because both α helices and β sheets have very specific sets of interactions of less than 5 Å between their hydrogen atoms (see Figure 18.19b). It is also possible to derive models of the three-dimensional structure of the protein molecule. However, usually a set of possible structures rather than a unique structure (Figure 18.20) is obtained, each of the possible structures obeying the distance constraints equally well. The sets of possible structures, which are frequently seen in NMR articles, do not, therefore, represent different actual conformations of a protein molecule present in solution. Rather they are simply the different structures that are compatible with data obtained by current methods. The primary source of this ambiguity is an insufficient number of measured distance constraints. Because of this ambiguity, the accuracy of an NMR structure is not constant over the whole molecule and is also difficult to quantify.

In addition to the problem of ambiguity, there are other limitations to the use of NMR methods for the determination of protein structures. The most severe concerns the size of the protein molecules whose structures can be determined. Currently, the upper limit is molecules with molecular weights of around 25 kDa, but this limit will be increased in the future by using improved methods and equipment. Furthermore, the method requires highly concentrated protein solutions, on the order of 1–2 mM, with the additional requirement that the protein molecules must not aggregate at these concentrations. In addition, the pH of the solution should be lower than about 6 for proton NMR experiments. The exchanges of the NH protons in the main chain become so fast at higher pH that it is very difficult to observe them with NMR, and the signals from these hydrogen atoms are essential for the sequential assignment procedure.

How well do NMR-derived structures agree with those determined by x-ray methods? The structures of some different globular proteins that have been independently obtained by the two methods—such as bovine pancreatic

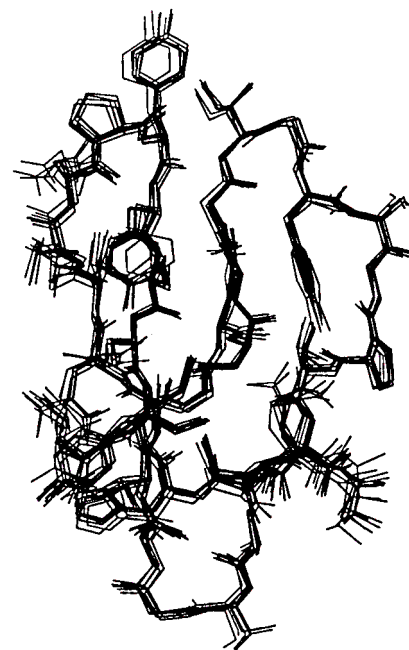


Figure 18.20 The two-dimensional NMR spectrum shown in Figure 18.17 was used to derive a number of distance constraints for different hydrogen atoms along the polypeptide chain of the C-terminal domain of a cellulase. The diagram shows 10 superimposed structures that all satisfy the distance constraints equally well. These structures are all quite similar since a large number of constraints were experimentally obtained. (Courtesy of P. Kraulis, Uppsala, from data published in P. Kraulis et al., *Biochemistry* 28: 7241–7257, 1989, by copyright permission of the American Chemical Society.)

trypsin inhibitor (see Figure 2.14a), plastocyanin (see Figure 2.11c) and thioredoxin from *E. coli* (see Figure 2.7)—show that NMR and x-ray crystallography give nearly identical results. The minor differences that exist are of the same order of magnitude as usually seen between x-ray structures of unrelated crystal forms of the same protein or determinations made under different experimental conditions. In other words, they are mostly small differences in loop regions of the main chain and different conformations of exposed side chains.

The situation is different for other examples—for example, the peptide hormone glucagon and a small peptide, metallothionein, which binds seven cadmium or zinc atoms. Here large discrepancies were found between the structures determined by x-ray diffraction and NMR methods. The differences in the case of glucagon can be attributed to genuine conformational variability under different experimental conditions, whereas the disagreement in the metallothionein case was later shown to be due to an incorrectly determined x-ray structure. A re-examination of the x-ray data of metallothionein gave a structure very similar to that determined by NMR.

NMR and x-ray crystallography are in many respects complementary. X-ray crystallography deals with the structure of proteins in the crystalline state, while NMR determines the structure in solution. The time scales of the measurements are different: NMR is more suitable for investigation of various dynamic processes such as those during folding, while x-ray crystallography is more suitable for characterization of protein surfaces and the water structure around the protein. X-ray crystallography remains the only method available to determine the structure of large protein molecules, whereas NMR is the method of choice for small protein molecules that might be difficult to crystallize.

Biochemical studies and molecular structure give complementary functional information

Our current knowledge of the relation between structure and function of protein molecules is insufficient to deduce the function of a protein from its structure alone, although, as we have seen, structural homology with proteins of known function can sometimes allow this. It is necessary to combine biochemical studies with structural information. Biochemical and cell biological studies can tell us if a protein is a receptor, a transport molecule, or an enzyme and, in addition, which ligands can bind to it, as well as the functional effects of such ligand binding. Studies of the three-dimensional structure of complexes between specific ligands and the protein will then give detailed information on how the active site is constructed and which amino acid residues are involved in ligand binding. Examples that we have described include protein–DNA interaction in Chapters 8, 9, and 10, sugar binding to a sugar transport protein in Chapter 4, and binding of inhibitors to enzymes that cleave peptide bonds in Chapter 11.

The specific role of each amino acid residue for the function of the protein can be tested by making specific mutations of the residue in question and examining the properties of the mutant protein. By combining in this way functional studies in solution, site-directed mutagenesis by recombinant DNA techniques, and three-dimensional structure determination, we are now in a position to gain fresh insights into the way protein molecules work.

Conclusion

The three-dimensional structure of protein molecules can be experimentally determined by two different methods, x-ray crystallography and NMR. The interaction of x-rays with electrons in molecules arranged in a crystal is used to obtain an electron-density map of the molecule, which can be interpreted in terms of an atomic model. Recent technical advances, such as powerful computers including graphics work stations, electronic area detectors, and

very strong x-ray sources from synchrotron radiation, have greatly facilitated the use of x-ray crystallography.

Crystallization of proteins can be difficult to achieve and usually requires many different experiments varying a number of parameters, such as pH, temperature, protein concentration, and the nature of solvent and precipitant. Protein crystals contain large channels and holes filled with solvents, which can be used for diffusion of heavy metals into the crystals. The addition of heavy metals is necessary for the phase determination of the diffracted beams.

X-ray structures are determined at different levels of resolution. At low resolution only the shape of the molecule is obtained, whereas at high resolution most atomic positions can be determined to a high degree of accuracy. At medium resolution the fold of the polypeptide chain is usually correctly revealed as well as the approximate positions of the side chains, including those at the active site. The quality of the final three-dimensional model of the protein depends on the resolution of the x-ray data and on the degree of refinement. In a highly refined structure, with an R value less than 0.20 at a resolution around 2.0 Å, the estimated errors in atomic positions are around 0.1 Å to 0.2 Å, provided the amino acid sequence is known.

Biological fibers, such as can be formed by DNA and fibrous proteins, may contain crystallites of highly ordered molecules whose structure can in principle be solved to atomic resolution by x-ray crystallography. In practice, however, these crystallites are rarely as ordered as true crystals, and in order to locate individual atoms it is necessary to introduce stereochemical constraints in the x-ray analysis so that the structure can be refined by molecular modeling.

In NMR the magnetic-spin properties of atomic nuclei within a molecule are used to obtain a list of distance constraints between those atoms in the molecule, from which a three-dimensional structure of the protein molecule can be obtained. The method does not require protein crystals and can be used on protein molecules in concentrated solutions. It is, however, restricted in its use to small protein molecules.

Selected readings

Bernstein, F.C., et al. The protein data bank: a computer-based archival file for macromolecular structures. *J. Mol. Biol.* 112: 535–542, 1977.

Blundell, T.L., Johnson, L.N. *Protein Crystallography*. London: Academic Press, 1976.

Branden, C.-I., Jones, A. Between objectivity and subjectivity. *Nature* 343: 687–689, 1990.

Clore, G.M., Gronenborn, A.M. Determination of three-dimensional structures of proteins and nucleic acids in solution by nuclear magnetic resonance spectroscopy. *CRC Crit. Rev. Biochem.* 24: 479–564, 1989.

Drenth, J. *Principles of protein x-ray crystallography*. Berlin: Springer-Verlag, 1994.

Eisenberg, D., Hill, C.P. Protein crystallography: more surprises ahead. *Trends Biochem. Sci.* 14: 260–264, 1989.

Ferre-D'Amare, A.R., Burley, S.K. Use of dynamic light scattering to assess crystallizability of macromolecules and macromolecular assemblies. *Structure* 2: 357–359, 1994.

Hendrickson, W. Determination of macromolecular structures from anomalous diffraction of synchrotron radiation. *Science* 254: 51–58, 1991.

Kleywegt, G.J., Jones, T.A. Where freedom is given, liberties are taken. *Structure* 3: 535–540, 1995.

McPherson, A. *The Preparation and Analysis of Protein Crystals*. New York: Wiley, 1982.

McPherson, A., et al. The science of macromolecular crystallization. *Structure* 3: 759–768, 1995.

Rodgers, D.W. Cryocrystallography. *Structure* 2: 1135–1140, 1994.

Walter, R.L., et al. High resolution macromolecular structure determination using CCD detectors and synchrotron radiation. *Structure* 3: 835–844, 1995.

Wilson, H.R. *Diffraction of X-rays by Proteins, Nucleic Acids and Viruses*. London: Edward Arnold, 1966.

Wright, P. What can two-dimensional NMR tell us about proteins? *Trends Biochem. Sci.* 14: 255–260, 1989.

Wüthrich, K. *NMR of Proteins and Nucleic Acids*. New York: Wiley, 1986.

Wüthrich, K. Protein structure determination in solution by nuclear magnetic resonance spectroscopy. *Science* 243: 45–50, 1989.

Wyckoff, H.W., Hirs, C.H.W., Timasheff, S.N. Diffraction methods for biological macromolecules. *Methods Enzymol.* 114: 330–386, 1985.

Protein Structure on the World Wide Web

The World Wide Web has transformed the way in which we obtain and analyze published information on proteins. What only a few years ago would take days or weeks and require the use of expensive computer workstations can now be achieved in a few minutes or hours using personal computers, both PCs and Macintosh, connected to the internet. The Web contains hundreds of sites of interest to molecular biologists, many of which are listed in **Pedro's BioMolecular Research Tools** (http://www.fmi.ch/biology/research_tools.html). Many sites provide free access to databases that make it very easy to obtain information on structurally related proteins, the amino acid sequences of homologous proteins, relevant literature references, medical information and metabolic pathways. This development has opened up new opportunities for even non-specialists to view and manipulate a structure of interest or to carry out amino-acid sequence comparisons, and one can now rapidly obtain an overview of a particular area of molecular biology. We shall here describe some Web sites that are of interest from a structural point of view. Updated links to these sites can be found in the *Introduction to Protein Structure* Web site (<http://www.ProteinStructure.com/>).

Many Web sites offer the opportunity to view the structures of proteins interactively, using a protein's atomic coordinates to produce images of different types that can be rotated and zoomed. Both the atomic coordinates and the computer software required for this can be freely either accessed from the Web or downloaded to desktop computers for off-line use. Three commonly used and free programs to view proteins structures on personal computers are RasMol, Chime and Mage. **RasMol** (<http://www.umass.edu/microbio/rasmol/>) runs on personal computers and produces interactive molecular images from a molecule's atomic coordinates. **Chime** (<http://www.mdli.com/tech/chemscape.html>) is a plug-in for Web browsers that allows interactive RasMol-like images to be embedded within Web pages. **Kinemages** (http://www.faseb.org/protein/kinemages/kin_page.html) are interactive molecular images produced by the program Mage. Many on-line journals use the kinemage format to display protein structures, and kinemages illustrating many of the protein structures discussed in this book can be found in the *Introduction to Protein Structure* Kinemage supplement disks (for further information, see <http://www.ProteinStructure.com>).

The **Brookhaven Protein Data Bank, PDB** (<http://www.pdb.bnl.gov>), is the primary store of experimentally determined atomic coordinates of proteins. Each coordinate set has a unique identification code that can be

retrieved together with the coordinates and other information about the structure, including interactive images, by searching for the protein's name or publication details. The ENTREZ search engine at the **National Center for Biotechnology Information** (NCBI; <http://www3.ncbi.nlm.nih.gov/Entrez/>) allows the searching of integrated databases for, amongst other things, protein sequence data, protein structures and bibliographic data. This site also provides the Vector Alignment Tool (VAST), allowing one to find and view similar structures, and the Basic Alignment Search Tool (BLAST), allowing one to find similar sequences.

There are also several databases that have arranged all known protein structures into some classification scheme. **SCOP** (<http://scop.mrc-lmb.cam.ac.uk/scop/>) is a database of structural domains arranged in a hierarchical manner according to structural and evolutionary relatedness. The SCOP site allows structures to be viewed interactively and enables the search for proteins with similar amino acid sequences using BLAST. **CATH** (<http://www.biochem.ucl.ac.uk/bsm/cath/>) arranges domains according to class (secondary structure composition), architecture (orientations of the secondary structures) and topology (shape and connectivity of secondary structure). The CATH site also includes a useful glossary of terms used in the description of protein structures. Protein topologies are extensively discussed in **TOPS** (<http://www3.ebi.ac.uk/tops/>), a site that searches for specific topologies and which can determine the topology of a new protein. **FSSP** (<http://www2.ebi.ac.uk/dali/fssp/>) is based on an all-against-all comparison of known structures. FSSP allows one to view similar structures superimposed and to obtain sequences of homologous proteins. A network service that allows the comparison of three-dimensional structures is **DALI** (<http://www2.ebi.ac.uk/dali/>). The coordinates of a new structure can be submitted to the site and DALI will check these coordinates against known structures to reveal biologically interesting similarities.

The ExPASy molecular biology Web site of the **Swiss Institute of Bioinformatics** (<http://www.expasy.ch>) covers many aspects of protein sequence and structure. It includes SWISS-PROT, which is an annotated protein sequence database, and SWISS-MODEL, an automated knowledge-based protein modeling server that allows one to model the three-dimensional structure of a protein whose sequence is known, based on the known structure of a homologous protein. Two very useful databases concerning the compilation and multiple sequence alignment of homologous domains are **Pfam** (<http://www.sanger.ac.uk/Pfam/>) and **ProDom** (<http://protein.toulouse.inra.fr/prodom.html>).

There are now a large number of Web sites devoted to particular fields of research. The **Nucleic Acid Database atlas** (<http://ndbserver.rutgers.edu/NDB/ndb.html>) provides a resource for viewing many DNA, RNA, protein-DNA and protein-RNA structures that have been solved by x-ray crystallography. In addition, many researchers produce Web pages that are concerned with their own particular research interests. As examples, an interesting site to learn about viruses is <http://www.bocklabs.wisc.edu/Welcome.html>, and an introduction to the disulfide bond-coupled folding pathway of bovine pancreatic trypsin inhibitor can be found at <http://www.biology.utah.edu/People/regfaculty/~goldenberg/GoldenbergLab/research/bpti.html>.

Index

- Accessory chlorophyll 238, 238F, 239
- Actin 197, **290–291**, 291F
 cross-bridges with myosin 291–292
 F-actin 293
 fibrous protein 292
 G-actin 293, 293F
 myosin complex structure 295, 295F
 structure 293–295
- Activation energy of reactions 206
 enzymes decreasing 206–207, 207F
- Active sites 3F
 α/β structures
 barrels 53–54, 53F
 domains 57, 59
 open twisted domains 56, 57, 57F, 59, 59F
 prediction 57, 59
 arabinose-binding protein 62F, 63
 carboxypeptidase 62F
 chymotrypsin 211–212, 211F, 212F
 crevices 57, 59, 63
 neuraminidase 71F, 72
 parallel β strands flanked by antiparallel β strands 62
 RuBisCo 53F
 serine protease 211–212, 211F, 212F, 361
 subtilisin 216–217, 216F
 tyrosyl-tRNA synthetase 59–60, 59F, 60F
- Acyl-enzyme intermediates 208, 208F
- Additivity principle 362
- Adenine
 bacteriophage MS2 RNA 340
 in DNA, hydrogen bonds 123F
- Adenylate cyclase, G protein-mediated
 activation 253, 253F
- Adenylate kinase 58F, 59
- ADP 115
- ADP–vanadate 295
- Alanine
 collagen mutation 285, 285F
 specificity of serine proteinases 213, 214F
 structure 6F
 substrate-assisted catalysis 218
- Alcohol dehydrogenase
 helical wheel 17T
 α helix 18
 zinc in 11, 11F
- Allosteric control 113, 142
- Allosteric effectors 142
see also trp repressor
- Allosteric proteins
 phosphofructokinase **114–117**
 switch between T and R states **113–114**
see also trp repressor
- Allotery 113
- $\alpha + \beta$ structures 32
 Cro protein 132
 lysozyme from T4 bacteriophage 355F
- α domains 31, **35–46**
 coiled-coil α helices 35–37
 doughnut-shaped 39–40, 39F
 evolutionary conservation 41–42
 four-helix bundle 37–39, 38F
 globin fold 22F, 35, 40, 40F
 helix movements and side-chain mutations 43
 hydrophobic interior 35, 42–43
 packing of α helices 40–41, 41F, 42F
 size and complexity 39–40
 tyrosyl-tRNA synthetase 59–60, 59F, 60F
- α helix (helices) **14–19**, 15F
 3_{10} helix 15, 17F
 6-F helix 116
 amino acids preferred 16–18
 in α/β structures
 barrels 49, 49F
 horseshoe folds 55, 56F
 twisted open sheets 56–57, 57F
- in β - α motif 27–28, 28F
- bacteriophage MS2 subunits 339
- barnase folding intermediate 94, 94F
- b/HLH/zip family 200
- buried and exposed types and sequences 17T
- C- and N-termini 16
- calmodulin 110
- CDK2 (PSTAIR helix) 107F, 108, 108F, 109F
- coiled-coil **35–37**, 35F, 36F
- GAL4 187
- heptad repeats 35–37, 36F, 192, 286
- leucine zipper dimerization 192
- oligomers of fibrous proteins 286–287
- stabilization by salt bridges 36, 37F
- conversion from β sheet (protein design) 368–370, 369F, 369T
- Cro protein 132
- dipole moment 16
 stabilizing to increase protein stability 357–358, 357F
- in α domains 31, 35
- DsbA 97
- electron-density maps 381F, 382
- F-actin 293
- four-helix bundle **37–39**, 38F
- GAL4 187
- G γ 263
- globin fold 40, 40F
- globular proteins 15
- GroEL domain 100, 102
- growth hormone 267, 267F
- handedness 16, 285
- helical wheel 17T
- in helix-turn-helix motif 129
- hemagglutinin 79
- homeodomains 160
- hydrogen bonds 15
- hydrophilic and hydrophobic residues 17
- lambda repressor 133
- length 15
- leucine zipper motif 192, 192F
- ligand-binding 16
- location in proteins 17
- loop region connections 21
- membrane proteins 223
- in membrane-bound proteins 35
- motifs using 24
- movements and side-chain mutations 43
- MyoD 197, 198–199
- myosin 294
- p53 169–170
 domain 167
- packing arrangements 40–41
- packing in α domains 40–41, 41F, 42F
- prediction of secondary structure 352
- proline in 16–17
- Ramachandran plot 10
- Ras protein 255
- recognition α helix 134
- residues per turn 15
- ridges and grooves geometry 40–41, 41F
- Rop protein 38–39, 39F
- schematic diagram representation 23
- secondary structure of proteins 14–16

Page numbers in **bold** refer to a major text discussion; page numbers with an F refer to a figure; page numbers with a T refer to a table.

- serpins 111–112
 SH2 domain 273
 side chains 17, 43
 subtilisin 215
 SV40 343, 343F
 TATA box-binding protein 154
 transducin G α 256, 257
 transmembrane *see* Transmembrane α helices
 transmembrane proteins 18, 223
trp repressor 142
 variations 15
 x-ray diffraction data 382
 in zinc motif of glucocorticoid receptor 184–185
- α/β barrels 44, **48–49**
 active sites 53–54, 53F
 prediction 57, 59
 amino acid residues 48, 50T
 branched hydrophobic side chains 49–51
 double 52–53, 52F
 enzymes containing 48–49
 evolution of new enzyme activities 54–55
 methylmalonyl-coenzyme A mutase 50–51, 50T
 parallel β strands 48, 49F, 306
 in pyruvate kinase 51–52, 51F
 α/β domains 32, **47–65**, 48F
 active site prediction 57, 59
 arabinose-binding protein 62–63, 62F
 α/β horseshoe folds 55, 56F
 α/β twisted open-sheet structures 47, **56–57**
 carboxypeptidase 60–62, 61F
 classes 47
 α domain with in tyrosyl-tRNA synthetase 59–60, 59F, 60F
 G-actin 293, 293F
 mixed β sheet with 60–62, 61F
 pyruvate kinase 51–52, 51F
 two similar in arabinose-binding protein 62–63, 62F
 tyrosyl-tRNA synthetase 59–60, 59F, 60F
 see also α/β barrels; α/β proteins
 α/β proteins/structures
 β helix relationship 84
 carboxypeptidase 60–62, 61F
 classes 47–48
 phosphofructokinase 115, 115F
 Ras protein 255
 subtilisin 215
 see also α/β domains
 α/β sheets
 open twisted 47, **56–57**
 active-site crevice 56–57, 57F, 63
 two domains in proteins 63
 types 56–57
- Alphaviruses **340–341**
 catalytic triad in coat protein protease 341
 core proteins 341, 341F
 Alzheimer's amyloid β -protein precursor inhibitor (APPI) 361, 362, 362T
 Alzheimer's disease 113, 283, 288
 Amide-hydrogen exchange 95
 Amino acid sequence 3F, 4
 assignment to three-dimensional folds 353–354
 calcium-binding motif 26F
 calculation of possible number 352
 γ -crystallin 76
 DNA binding domain of glucocorticoid receptor 182F
 Fos and Jun 192F
 GCN4 192F, 193, 194F
 heptad, in coiled-coil α helix 35–37, 36F
 HLH region 201F
- homeodomains 160, 162, 162F
 homologous from different species 21
 homologous proteins 348
 in interpretation of NMR spectra 390
 Max protein 192F
 photosynthetic reaction center 247F
 protein structure prediction from 352–353
 retinol-binding protein (RBP) 69–70, 70F
 Rop 369T
 similar three-dimensional structures 352
 TATA box-binding protein 153
 transmembrane α helices prediction 244–245
 Zif 268 protein 177F, 368T
- Amino acids
 in α/β barrels 48, 50T
 basic, in helix-loop-helix region 196
 classes 5
 D- and L- forms 5, 9
 handedness 5
 nomenclature 7F
 preferred in α helix 16–18
 role of individual residues 391
 side chains *see* Side chains
 structures 4, 4F
 Amino groups 4, 4F
 Aminoacyl-tRNA synthetases, domains 59–60, 59F, 60F
 Ammonium ions 211, 213
 Amplitude, diffracted beams 370F, 379, 380, 380F
 β Amyloid 290
 Amyloid β -protein precursor inhibitor, Alzheimer's (APPI) 361, 362, 362T
 Amyloid fibers 289, 297
 Amyloid fibrils 288–289
 Amyloidosis 288
 Antenna pigment proteins 235, 240–241
 ring in LH1 light-harvesting complex 242–244, 243F
Antennapedia 159, 160, 162, 162F
 Antibodies 299
 antigen-binding sites *see* Antigen-binding sites
 catalytic 207, 309
 production 309
 chimeric 306
 diversity 302–303
 membrane protein solubilization 224–225
 polypeptide chains 300–301
 see also Immunoglobulin(s)
 Antigen, recognition in MHC molecules 314–315, 315F, 316
 Antigen-binding sites 21
 for antigens 308–311
 for haptens 308–309, 309F
 of immunoglobulins 21, 306–308, 306F, 308–311
 modeling 349–350
 for lysozyme 309–310, 310F
 of MHC class I molecule 314
 of T-cell receptors (TCR) 317F
 Antigenic determinants 299
 Antigen-presenting cells 315
 Antiparallel β barrels
 chymotrypsin 210–211, 210F
 cyclophilin 99, 99F
 GroES 102
 p53 DNA-binding domain 168–169, 168F
 porins 229–230, 230F
 viruses 335, 335F
 Antiparallel β hairpin *see* Hairpin β motif
 Antiparallel β sandwich, TFIIA 159
 Antiparallel β sheets 18F, 19, 67, 67F
 in β domains 31
 bacteriophage MS2 subunits 339
- DNA-binding site of TATA box-binding protein 154
 Greek key motifs 27
 immunoglobulin fold 304–305, 304F
 lambda Cro protein 132
 MHC class I molecule 314
 neuraminidase 71–72, 71F
 p53 domain 167
 serpin fold 111
 SH2 domain 273
 topology diagram 24F
 Antiparallel β strands
 active site of carboxypeptidase 62
 GroEL 102
 jelly roll barrel formation 77–78, 77F
 in up-and-down β barrel 68F, 69–70, 70F
 variable domain of immunoglobulin 306, 308
 Antiparallel β structures **67–84**
 γ -crystallin 74, 74F
 Greek key motifs 72–74
 see also Greek key motif proteins
 included 67
 SH3 domain 274, 275F
 up-and-down β barrels (sheets) *see* Up-and-down β barrels; Up-and-down β sheets
 Antithrombin 111, 112, 113
 α_1 -Antitrypsin 110–111
 active, cleaved and latent forms 112, 112F
 deficiency 113
 Antiviral compounds, design for common cold 337–338, 337F, 338F
 AP1 complex 192
 pseudo-symmetric binding site 194, 194F, 196
 Aphthoviruses 333
 APPI (Alzheimer's amyloid β -protein precursor inhibitor) 361, 362, 362T
Arabidopsis thaliana 154
 Arabinose 63F
 transport 62
 Arabinose-binding protein, α/β domains 62–63, 62F
 Area detectors 377
 Arginine
 p53 binding to DNA 170, 170F, 171
 recognition helix of glucocorticoid receptor 184–185
 structure 7F
 zinc finger interaction with DNA 179, 179F, 181
 Aromatic residues, hydrophobicity in porins 231
 Asparagine
 structure 6F
 TATA box and TBP binding 157–158
 Aspartate transcarbamoylase 24F
 Aspartic acid
 in G β subunit 263
 mutation in trypsin 215
 in parvalbumin calcium-binding motif 25
 in serine proteinases 209
 structure 6F
 substrate-assisted catalysis 218
 Aspartic proteinases 205
 Asymmetric units
 definition 328
 icosahedron 328, 328F
 A-T base pair 123F
 TATA box 158
 AT sequence, p53 binding to DNA 170
 ATP
 binding to phosphofructokinase 116
 cyclin A and CDK2 binding 108, 109F
 depletion in rigor mortis 295, 296
 GroEL–GroES complex 101F, 103